

DePARylation is critical for S phase progression and cell survival

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Abstract

Poly(ADP-ribose)ylation or PARylation by PAR polymerase 1 (PARP1) and dePARylation by poly(ADP-ribose) glycohydrolase (PARG) are equally important for the dynamic regulation of DNA damage response. PARG, the most active dePARylation enzyme, is recruited to sites of DNA damage via pADPr-dependent and PCNA-dependent mechanisms. Targeting dePARylation is considered an alternative strategy to overcome PARP inhibitor resistance. However, precisely how dePARylation functions in normal unperturbed cells remains elusive. To address this challenge, we conducted multiple CRISPR screens and revealed that dePARylation of S phase pADPr by PARG is essential for cell viability. Loss of dePARylation activity initially induced S phase-specific pADPr signaling, which resulted from unligated Okazaki fragments and eventually led to uncontrolled pADPr accumulation and PARP1/2-dependent cytotoxicity. Moreover, we demonstrated that proteins involved in Okazaki fragment ligation and/or base excision repair regulate pADPr signaling and cell death induced by PARG inhibition. In addition, we determined that PARG expression is critical for cellular sensitivity to PARG inhibition. Additionally, we revealed that PARG is essential for cell survival by suppressing pADPr. Collectively, our data not only identify an essential role for PARG in normal proliferating cells but also provide a potential biomarker for the further development of PARG inhibitors in cancer therapy.

Significance statement

Poly(ADP-ribosyl)ation is a reversible post-translational modification. Although PARG may have a protective effect against excessive PARP1 engagement, detailed knowledge of PARG's mechanism of action remains elusive. Here, we showed that PARG participates in DNA replication, especially in Okazaki fragment maturation. Moreover, PARG level is critically important for cellular sensitivity to PARG inhibition, which is a valuable biomarker for PARGi-based therapy.

eLife assessment

The demonstration that the PARG dePARylation enzyme is required in S phase to remove polyADP-ribose (PAR) protein adducts that are generated in response to the presence of unligated Okazaki fragments is potentially **valuable**, but the evidence is **incomplete**, and identification of relevant PARylated PARG substrates in S-phase is needed to understand the role of PARP1-mediated PARylation and PARG-catalyzed dePARylation in S-phase progression.

Introduction

Poly(ADP-ribosyl)ation or PARylation is a conserved post-translational modification that is important for many cellular processes, including DNA damage repair^{1,2}. PARylation is mainly driven by poly(ADP-ribose) polymerase 1 (PARP1), and to a lesser extent by PARP2, using NAD⁺ as an ADP-ribose donor and generating nicotinamide (NAM) as a byproduct in human cells. Extensive studies have already established the key roles of PARP1/2 in DNA damage response (DDR)^{3,4}. As a DNA damage sensor, PARP1 rapidly recognizes and binds to DNA damage sites, which dramatically activates its own enzymatic activity and thereby modifies itself and other proteins with pADPr and mono-(ADP-ribosylation). These modifications lead to the recruitment of proteins that are involved in DDR and DNA repair. The binding to DNA by activated PARP1 acts as a “hit and run” mechanism⁵, which restricts PARylation at a specific point and facilitates the subsequent engagement of other repair proteins. Thus, PARylation by PARP1, when responding to DNA damage, is a rapid and transient process. In addition, PARP1 can be activated by unligated Okazaki fragments to induce endogenous S phase pADPr, which then recruits XRCC1 and LIG3 to facilitate the ligation of these Okazaki fragments⁶.

PARylation is a reversible post-translational modification. The removal of pADPr is mainly carried out by poly(ADP-ribose) glycohydrolase (PARG)⁷. However, PARG cannot remove the terminal ADP-ribose, whose removal requires additional hydrolases, including terminal ADP-ribose glycohydrolase (TARG1), ADP-ribose-acceptor hydrolases ARH1/3, and possibly other macrodomain-containing proteins, such as hMacroD1/D2⁸. Later studies showed that PARG and ARH3 (ADPRHL2/ADPRS) are the primary dePARylation enzymes in vertebrates, although ARH3 has much lower activity against pADPr than PARG and mainly functions as a serine-directed mono-ADP-ribosylhydrolase^{9–11}. As an endo-glycohydrolase and exo-glycohydrolase, PARG specifically hydrolyzes the glycosidic bonds. PARG may function as an oncogene, as the high level of PARG promotes cell transformation and invasion and is associated with poor overall survival¹². Moreover, complete loss of PARG leads to embryonic lethality in mice¹³. Importantly, PARG is recruited to sites of DNA damage by PARP1- or pADPr-dependent and PCNA-dependent mechanisms¹⁴ and involved in both DNA double-strand breaks (DSBs) and single-strand breaks (SSBs) repair. Loss of PARG sensitizes cells to DNA damage agents^{15–17}. In addition, PARG is also involved in transactions at replication forks^{18,19}.

On average, a single cell suffers thousands of DNA damage events every day²⁰; PARylation and dePARylation occur rapidly and collaboratively to facilitate DNA repair in the cell. This rapid PARylation and dePARylation cycle is likely important for homeostasis regulation. Unrestrained PARylation following DNA damage may lead to a significant change in the NAD⁺ level in cells, which not only inhibits other cellular processes that require NAD⁺ but also results in cell death due to NAD⁺ depletion and/or PARP1-dependent cell death called Parthanatos^{10,21}. Thus, PARG and other dePARylation enzymes are critically important for the recycling of NAD⁺ and the control of NAD⁺ homeostasis^{22–24}.

Targeting PARylation and dePARylation is a promising strategy in cancer and other therapies. PARP inhibitors (PARPis) have already been approved for clinical treatment of cancers with homologous recombination (HR) deficiency, while targeting dePARylation is an encouraging alternative strategy to overcome PARPi resistance^{25–29}. Given the critical roles of PARG in the maintenance of the PARylation/dePARylation cycle in cells, PARG inhibitors have been developed recently as potential anti-cancer agents²⁸. These include PDD00017273 (PARGi) (**Fig. S1A**)³⁰, COH34²⁶, and JA2131³¹. Interestingly, cells with deficiency in HR and other DNA damage signaling/repair pathways showed increased sensitivity to the PARG inhibitors PDD00017273 and COH34^{25, 26}. These PARG inhibitors could also sensitize cells to radiation therapy and chemotherapy^{23, 26, 31, 32}. Thus, there is significant interest in further developing these PARG inhibitors for cancer treatment.

Despite the crucial role of PARG in dePARylation in multiple pathways, detailed knowledge of its mechanism of action remains elusive. Especially, how PARG regulates dePARylation in normal unperturbed cells and the balance between PARylation and dePARylation remain unclear. To address this issue, we performed multiple CRISPR screens and created PARP1/2 DKO, PARG KO, and additional KO cells. We showed that loss of dePARylation activity induced S phase-specific pADPr signaling, which likely originated from unligated Okazaki fragments. Consequently, the failure to remove S phase pADPr led to PARylation- or PARP1/2-dependent cell death. Furthermore, perturbation of Okazaki fragment ligation and/or base excision repair (BER) increased pADPr signaling and promoted cell death induced by PARGi. In addition, we revealed that the PARG level is a potential biomarker for PARGi-based cancer therapy. Moreover, we showed that PARG is essential for cell survival, which can be exploited for cancer treatment.

Materials and Methods

Cell culture

HEK293A, HeLa, and OVCAR3 cells were purchased from the American Type Culture Collection (ATCC); HEK293A and HeLa cells were cultured in DMEM with 10% fetal bovine serum, while OVCAR3 cells were cultured in RPMI with 10% fetal bovine serum. RPE1-hTERT FLAG-Cas9 TP53^{-/-} BRCA1^{-/-} cells were a gift that was kindly provided by Dr. Daniel Durocher (University of Toronto) and were cultured in DMEM with 10% fetal bovine serum. RMUGS was obtained from the Japanese Collection of Research Bioresources Cell Bank and cultured in Ham's F12 medium with 10% fetal bovine serum. Knockout cells generated in 293A and HeLa cells were created with pLentiCRISPRv2 (Addgene, #52961) containing indicated gRNAs (**Fig. S8**), as described previously³³. All knockout cells were validated by Western blotting and DNA sequencing (**Fig. S8**). The 293A-derived PARP1/2 DKO cells were the same as those in the previous study³³. All cell lines were free of mycoplasma contamination.

Chemical reagents and antibodies

The NAD/NADH Assay Kit II (colorimetric) (ab221821) was obtained from Abcam. The CellTiter-Glo Luminescent Cell Viability Assay (G7573) was purchased from Promega. Nicotinamide (NAM, S1899), β -nicotinamide mononucleotide (NMN, S5259), olaparib (S1060), and FK866 (S2799) were purchased from SelleckChem. PDD 00017272 (HY-133531), PDD 00017238 (HY-133530), PDD 00031705 (HY-135846), and PDD 00017273 (HY-108360) were obtained from MedChemExpress. Methyl methanesulfonate (MMS, 129925), thymidine (T9250), emetine (E2375), and crystal violet solution (HT90132) were from Sigma. The propidium iodide (PI, P3566), GATEWAY cloning system (11789100, 11791100) and FxCycle Violet Stain (F10347) were obtained from ThermoFisher. The QuikChange II Site-Directed Mutagenesis Kit (200523) was from Agilent. The CometAssay Single Cell Gel Electrophoresis Assay (4250-050-K) was obtained from Trevigen.

Antibodies against PARP1 (9532S), PARG (66564S, which mainly recognizes full-length isoform 1, but also weakly recognizes isoforms 2 and 3 (data not shown)), and XRCC1 (2735S) were purchased from Cell Signaling Technology. Antibodies against C-terminus of PARG were purchased from Santa Cruz Biotechnology (SC-398563) and Novus Biologicals (NBP2-55661). Antibodies against histone H3 (ab18521), γ H2AX (ab2893), and POLB (ab26343) were from Abcam. Antibodies against β -actin (A5316), tubulin (T6199), and ARH3 (HPA027104) were from Sigma. The antibodies anti-pADPr (10H) (sc-56198), LIG1 (sc-56087), and BRCA1 (sc-6954) were from Santa Cruz Biotechnology. Anti-PARP2 antibody (39743) was from Creative Motif, anti-LIG3 antibody (GTX70143) was from GeneTex, and anti-53BP1 antibody (NB100-304H) was from Novus Biologicals. Anti- γ H2AX (05-6361) and PAR (10H) (AM80) antibodies were obtained from Millipore. Alexa Fluor Plus 488 secondary antibody goat anti-mouse (A327230) was from ThermoFisher.

Plasmid constructs

PARP1 (HsCD00043719) and PARG (HsCD00859023) cDNAs, purchased from DNASU, were subjected to mutagenesis using the QuikChange II Site-Directed Mutagenesis Kit. WT and mutant constructs were subsequently cloned into the vector pLenti CMV Neo DEST (705-1) (Addgene, #17392) by the GATEWAY cloning system. shRNA constructs (pGIPZ-based vector) targeting PARG (Clone ID: V2LHS_11965, V2LHS_250448, V3LHS_379835, and V3LHS_379832) and targeting POLB (Clone ID: V2LHS_170201 and V2LHS_222459) were obtained from Horizon Discovery Biosciences. The Toronto KnockOut Library v3 (TKOv3) (90294) was from Addgene. The DNA Damage Response MKOv4 Library (Addgene, #140219) was from our previous study³⁴.

Cell viability assays

To measure cell viability, ~2,000 cells were seeded into the 96-well plates. Drugs with indicated concentrations (three biological replicates) were added after 24 hours. After 72 hours of treatment, cell numbers were measured by the CellTiter-Glo Luminescent Cell Viability Assay according to the manufacturer's instructions to draw the cell viability curves.

NAD⁺ level measurement

The NAD⁺ relative level was measured following the manufacturer's instructions. In brief, the same number of cells with the indicated treatment were collected and lysed with NAD⁺ extraction solution (0.5 M perchloric acid) for 30 min on ice. The extraction was neutralized by 0.55 M K₂CO₃ and centrifuged. The supernatant was collected and added to a 96-well plate. NAD⁺ standard and the NAD⁺ reaction mixture were added to the 96-well plate for the next colorimetric reading at OD 450 nm after a 30-min reaction. For each time point, at least three biological replicates were measured.

Clonogenic assay

To assess cellular sensitivity to PARGi or other agents, the indicated cells were seeded on 12-well plates and subsequently exposed to DMSO or indicated treatments for 7 to 14 days. After phosphate-buffered saline (PBS) washes, cells were stained with crystal violet solution. For each condition, three biological replicates were included and the representative results were presented.

Western blotting analysis

Unless further operation was indicated, such as chromatin and soluble fractionation, cells were washed with PBS and directly lysed on the plate by 2× Laemmli buffer, boiled at 95 °C for 10 min, and separated by sodium dodecyl sulfate polyacrylamide gel electrophoresis, transferred to membranes, and immunoblotted with the indicated antibodies. For each condition, at least two biological replicates were conducted and the representative results were presented.

Fluorescence-activated single cell sorting analysis (FACS)

The same number of trypsinized cells with the indicated treatments were collected into a 15-ml tube with PBS, which were subsequently fixed by pre-cold ethanol to a final 70% concentration. After centrifugation, cell pellets were permeabilized by PBS with 0.5% Triton X-100 at ambient temperature for 20 min. After being washed with PBS, cell pellets were blocked by PBS with 4% bovine serum albumin at ambient temperature for 1 hour. The cell pellets were incubated at 4 °C overnight with antibodies (pADPr, AM80; γH2AX, 05-6361) at 1:1000 dilution by PBS with 4% BSA. After being washed with PBS three times, cell pellets were incubated with an Alexa Fluor 488 secondary antibody at 1:500 dilution for another 1 hour at ambient temperature. The pellets were then stained with FxCycle Violet Stain (1 μg/ml) or PI (20 μg/ml) with RNase A (10 μg/ml) and resuspended in PBS for the next flow cytometry analysis by a BD C6 flow cytometer (Becton Dickinson) or Attune Flow cytometers (ThermoFisher). FlowJo software (v10.6.1) was used to analyze the acquired data. For each condition, at least three biological replicates were conducted and the representative results were presented.

Chromatin and soluble fractionation and enrichment for PARylated proteins

Cells with the indicated treatments were collected for chromatin and soluble fractionation with a protocol reported previously^{35, 36}. In brief, cells were lysed in NETN buffer (20 mM Tris-HCl [pH 8.0], 1 mM EDTA, 100 mM NaCl, 0.5% NP-40, and 1 mM DTT) containing proteinase inhibitors for 20 min on ice. After centrifugation, the soluble fraction (i.e. the supernatant) was collected into fresh tubes. The chromatin fractionation (i.e. the pellets) was washed twice with NETN buffer. The soluble and chromatin fraction were both diluted with 2× Laemmli buffer, boiled at 95 °C for 10 min, and subjected to Western blotting analysis. Three biological replicates of chromatin and soluble fractionation were conducted and the representative results were presented.

For enrichment of PARylated proteins in chromatin and soluble fractions, cells were lysed for 20 min on ice by NETN buffer containing proteinase inhibitors and PARP/PARGis (10 μM) to avoid PARylation/dePARylation during cell lysis. After fractionation, the chromatin fractionation (i.e. the pellets) was resuspended by NETN buffer containing proteinase inhibitors and PARP/PARGis (10 μM) and subjected to sonication and centrifugation. The chromatin and soluble fractions were incubated with Af1521 macrodomain affinity resin for 4 hours at 4 °C. After being washed three times with NETN buffer, the beads were boiled with 1× Laemmli buffer at 95 °C for 10 min. The elution was analyzed by Western blotting with the indicated antibodies. Two biological replicates of enrichment were conducted and the representative results were presented.

shRNA knockdown

pGIPZ vectors containing scrambled or gene-specific shRNAs were packed into lenti-virus with the packaging vector psPAX2, the envelope vector pMD2.G, and polyethyleneimine; then lenti-virus was used to infect indicated cell lines. Infected cells were selected with puromycin for ~5 days. The pooled cells were validated by Western blotting and used for further experiments.

Cell viability-based and flow cytometry-based CRISPR screens

DDR library screening was conducted as described previously³⁴. In brief, cells were infected with DDR library virus at a low multiplicity of infection (MOI) (~ 0.3) for 24 hours and then selected with puromycin (2 μg/ml) for 72 hours. Cells were collected as the initial time point T0, and the remaining cells (five million cells in each replicate, each condition containing at least two replicates) were passaged every 3 days for 21 days. Five million cells were collected at both T0 and the final time point at 21 days (T21).

For whole-genome CRISPR gRNA screening, 120 million cells were infected with lentiviruses encoding the TKOv3 library at a low MOI ratio (< 0.3) for 24 hours, as conducted previously³⁶. Infected cells were selected with puromycin (2 $\mu\text{g}/\text{ml}$) for 72 hours. For the cell viability-based screens, 20 million cells were collected after selection and marked as the initial time point (T0). The remaining cells (20 million cells in each replicate, with each condition containing at least two replicates) were passaged every 3 days for 21 days and treated with PARGi or DMSO. Twenty million cells were collected in each condition after 21 days and marked as T21. For flow cytometry-based screening, a similar workflow was performed as described previously³⁷. At day 5 after selection, 150 million cells in each replicate were treated with either PARGi (10 μM) for 4 hours or DMSO. Cells were subjected to sequential fixation, permeabilization, blocking, antibody incubation, and staining, as described above. After that, ~ 20 million cells were collected in both the top 25% with the highest signal and the bottom 25% with the lowest signal cell populations on the basis of the pADPr signal, as determined by flow cytometry.

Genomic DNA was extracted from cell pellets by the QIAamp Blood Maxi Kit (Qiagen) and resuspended in Buffer EB (10 mM Tris-HCl [pH 7.5]) after precipitation by ethanol and sodium chloride. DNA was then amplified by PCR with primers harboring Illumina TruSeq adapters with i5 and i7 barcodes, and the resulting libraries were sequenced on an Illumina NextSeq 500 system. The BAGEL algorithm (<https://github.com/hart-lab/bagel>) was used to calculate essentiality scores. A DrugZ analysis (<https://github.com/hart-lab/drugz>) was used to calculate the difference between different groups.

Alkaline comet assay

An alkaline comet assay was performed using a CometAssay kit under the manufacturer's instructions. In brief, equal numbers of cells from different treatments were collected at the same time and resuspended by PBS with a concentration of 1×10^5 cells/ml. The cell suspension (50 μl) was well-mixed with 500 μl of molten LMAgarose (at 37 $^{\circ}\text{C}$) and then immediately spread onto CometSlide. After solidification at 4 $^{\circ}\text{C}$ in the dark for 10 min, slides were incubated with lysis solution for 60 mins at 4 $^{\circ}\text{C}$ in the dark and then immersed into the alkaline unwinding solution (200 mM NaOH, 1 mM EDTA, pH >13) for 60 min at 4 $^{\circ}\text{C}$ in the dark. After that, slides were placed into an electrophoresis slide tray with the alkaline unwinding solution for 45 min at 23 voltage. Slides were gently washed, twice with ddH₂O and once with 70% ethanol. Slides were dried at 37 $^{\circ}\text{C}$ for ~ 20 min to make all cells in a single plane and stained with SYBR-Gold. Images were captured by a Nikon 90i microscope at $\times 10$ magnification and analyzed by OpenComet (V1.3.1).

CRISPR/Cas9-mediated stable knockout cells and reconstitution

Cells were transfected with pLentiCRISPRv2 plasmids containing the gRNAs targeting indicated genes (Fig. S8) using polyethyleneimine. After 24 hours, cells were selected with puromycin for another 48 hours and seeded into 96-well plates with 1 cell in each well. After 2 weeks, the single clones were selected from 96-well plates for further validation by Western blotting and DNA sequencing (Fig. S8). Each knockout clone was generated with single gRNA.

pLenti CMV Neo DEST expression vectors containing empty or indicated gene constructor were packed into lenti-virus with the packaging vector psPAX2, the envelope vector pMD2.G, and polyethyleneimine. Cells were infected with the indicated virus and selected with puromycin for ~ 5 days, after 24 hours. The pooled cells were confirmed with Western blotting to validate the expression of genes and used in further experiments.

The generation of PARG complete/conditional knockout (cKO) cells

Cells were transfected with pLentiCRISPRv2 plasmids containing the gRNAs targeting the indicated sequences (Fig. 7A) with polyethyleneimine. After selection with puromycin, cells were seeded into 96-well plates with 1 cell each well in the presence of low concentration of olaparib (100 nM).

Each cKO clone was generated with single gRNA. Viable clones were cultured with low concentration of olaparib and subjected to further validation by Western blotting and DNA sequencing (Fig. S9). To keep PARG cKO cells viable, validated clones were always cultured with the addition of low concentration of olaparib.

Double thymidine block

After cells had adhered to tissue culture plates, they were treated with 2 mM thymidine for 16 hours. The medium was removed, and the cells were washed with sterile PBS twice. Fresh medium was added for another 9 hours. After that, cells were incubated with thymidine for an additional 14 hours.

Human tissue IHC analysis

Human ovary and breast carcinoma tissue microarrays were purchased from US Biomax. The ovary carcinoma tissue (BC11115d) contains 5 cases of clear cell carcinoma, 62 serous carcinoma, 10 mucinous adenocarcinoma, 3 endometrioid adenocarcinoma, 10 lymph node metastasis carcinoma, 10 adjacent normal ovary tissue (missing one case adjacent normal ovary tissue and one case of serous carcinoma, so we have 98 cases in total for analysis). The breast carcinoma tissue (BC081120f) contains 100 cases of invasive carcinoma of no special type, 10 adjacent normal breast tissue (missing two cases of invasive carcinoma, so we have 108 cases in total for analysis). After deparaffinization and rehydration, antigen retrieval was done by Tris-EDTA Buffer (pH 9.0). The samples were treated with methanol with 1% hydrogen peroxide for 30 min to block endogenous peroxidase activity, followed by incubation with 10% normal goat serum to prevent nonspecific staining; then, the samples were incubated with anti-PARG antibody (1:50. Cell signaling Technology, 24489S) at 4 °C overnight. The samples were incubated with a biotinylated secondary antibody (1:200. Vector Laboratories, PK-6101) for 30 min at ambient temperature and with avidin-biotin peroxidase complex solution (1:100) for additional 30 min at ambient temperature. The slides were incubated with DAB solution and subsequently counterstained with haematoxylin. The slides were scanned on the Vectra Polaris Automated Quantitative Pathology Imaging System (PerkinElmer, Waltham, US) for quantification by Visiopharm platform (Visiopharm, Hoersholm, Denmark). A total score of protein expression was calculated from both the percentage of immunopositive cells and the immunostaining intensity. High and low protein expressions were defined using the mean score of all normal tissues as a cutoff point.

DePARylation activity of PARG in whole cell lysates

The dePARylation activity of PARG was measured in the PARP1 Histone H4 Activity Assay (#K611, Tulip BioLabs). Briefly, wells of the plate were incubated with activated PARP1 mixtures to generate PARylated Histone H4 or PARP1 mixtures without NAD^+ , which were set as the Blank, according to the manufacturer instruction. After that, the plate was washed with PBS three times. Equal number of cells prepared from different cell lines were lysed by equal amount of NETN buffer in the presence of 10 μM olaparib, to inhibit the endogenous PARP1 activity, on ice for 30 min. After centrifugation at 14,300 $\times g$ at 4 °C for 10 min, supernatant was kept, and the protein concentration was determined by BCA assay. Wells of the plate were either incubated with equal amount of whole cell lysates in NETN buffer with 10 μM olaparib or just NETN buffer with 10 μM olaparib, set as the Control, overnight at room temperature. The Blank was also incubated with equal NETN buffer with 10 μM olaparib. After that, the plate was washed with PBS three times. The remnant pADPr were detected by anti-pADPr antibodies and quantified by TMB substrate with the OD_{450} read, according to the manufacturer instruction. The percentage of pADPr hydrolysis was calculated by normalization with the Blank and the Control.

Results

PARG depletion leads to drastic sensitivity to PARGi

The results of early studies indicated that defective DNA repair pathways and DNA replication stress would result in enhanced sensitivity to PARG inhibitor, PDD00017273 (PARGi)^{25,27,38,39}. To further define the key DDR pathways that are important for cellular response to PARGi, we performed CRISPR screening using our homemade DDR sgRNA library, which targets approximately 360 genes involved in various DDR pathways³⁴. Interestingly, our screen results showed that PARG depletion led to significantly increased cellular sensitivity to PARGi (**Fig. 1A**, **Table S1**). In addition, we found that POLB (**Fig. 1A**), which is important for BER, a repair pathway that relies on PARP1 functions and pADPr-dependent DNA damage signaling, also showed synthetic lethality with PARGi. The synthetic lethality between POLB and PARGi was consistent with the findings of a previous report³⁸, indicating the high quality of our screening results.

To further validate our data, we generated PARG KO cells in the same 293A cells used in DDR sgRNA library screening. Interestingly, PARG KO cells showed extreme sensitivity to PARGi, with nearly thousand-fold sensitivity (the IC₅₀ in 293A cells was 96±24 μM and 210±30 nM in PARG KO cells, on the basis of five independent biological replicates) (**Fig. 1B**). In addition, we used another two structurally similar compounds (PDD00017272 and PDD00017238) to further confirm the extreme sensitivity of PARG KO cells to PARG inhibition (**Fig. S1B**), while the inactive but structurally similar compound PDD0031705 did not distinguish wild-type (WT) or PARG KO cells (**Fig. S1C**). To further validate that this sensitivity was not restricted to one cell line, we generated HeLa-derived PARG KO cells with a different gRNA. As expected, HeLa-derived PARG KO cells also displayed extreme sensitivity to PARGi (**Fig. S1D**). These results suggest that PARG depletion resulted in drastic sensitivity to PARGi.

We reconstituted PARG KO cells with WT PARG or a PARG catalytic-inactive mutant E755/756A⁴⁰ to further investigate the nature of PARGi sensitivity in PARG KO cells. Cells treated with the alkylating agent methyl methanesulfonate (MMS), which is known to activate pADPr signaling, were used to test PARG activity. As shown in **Fig. S1E**, PARG KO cells and cells reconstituted with PARG catalytically inactive mutant showed a significant increase in pADPr signaling induced by MMS, while WT cells or cells reconstituted with WT PARG did not. Expectedly, reconstitution with WT PARG reversed the sensitivity of PARG KO cells to PARGi, while reconstitution with PARG catalytic inactive mutant failed to do so (**Fig. 1C**). Together, these data strongly suggest that loss of PARG activity induces dramatic sensitivity to PARGi.

The question is why PARG KO cells would be sensitive to PARGi. There are at least two potential explanations for this observation. First, PARGi may target not only PARG but also a second target, which only becomes essential in the absence of PARG. Second, our PARG KO cells, despite being generated with two different sgRNAs and in two independent cell lines validated by Western blotting and DNA sequencing, are not complete PARG KO cells. The two different sgRNAs used here target all three catalytically active isoforms (isoforms 1, 2 and 3), and sgRNA#2 used in HeLa cells also targets isoforms 4 and 5, but these two isoforms are considered catalytically inactive according to the Uniprot database (**Fig. S1D**). To address this important question, we performed several experiments to test both hypotheses. Please see below for additional studies.

PARGi induces PARP1/2-dependent cell death in PARG KO cells

As shown above (**Fig. S1E**), MMS treatment significantly increased pADPr signaling in PARG KO cells. Interestingly, pADPr signaling further increased modestly in PARG KO cells treated with the combination of PARGi and MMS (**Fig. S2A**). To further evaluate the underlying mechanisms of PARGi sensitivity in PARG KO cells, we generated PARP1/2 DKO cells in both WT and PARG KO cells and treated them with PARGi (**Fig. 2A**). PARP1 cleavage was detected in PARG KO cells treated with PARGi, indicating apoptosis. Moreover, pADPr accumulated significantly in PARG KO cells

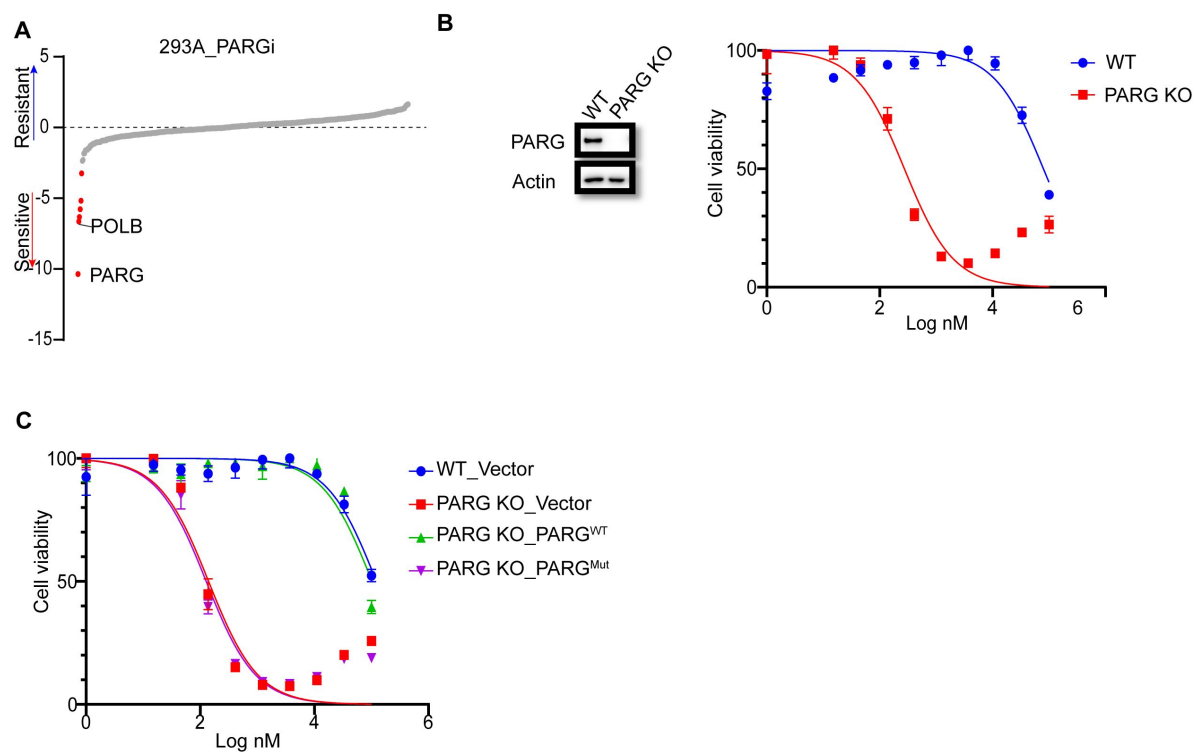


Fig. 1.

PARG loss sensitizes cells to PARGi.

a, Ranking of co-essential genes with PARGi treatment on the basis of a DrugZ analysis of the results of CRISPR/Cas9 screening with a DDR library in HEK293A cells. The NormZ score was used to determine a possible synthetic lethality gene under PARGi treatment. Drug-sensitive genes were marked in red; drug-resistant genes were marked in blue on the basis of the false discovery rate (FDR, 0.05 cut-off). **b**, HEK293A WT cells and HEK293A PARG KO cells were treated with different doses of PARGi for 72 hours. Cell viability was determined by the CellTiter-Glo assay. **c**, HEK293A PARG KO cells, re-constituted with either full-length PARG or catalytic domain mutation of PARG, were treated with different doses of PARGi for 72 hours. Cell viability was determined by the CellTiter-Glo assay.

treated with PARGi, but not in other cells (**Fig. S2B**). Furthermore, we used flow cytometry to detect pADPr signaling following short treatment with PARGi (**Fig. 2B**). Consistently, moderate pADPr signaling was observed in PARG KO cells, but not in other cells (**Fig. 2B**).

We reason that sensitivity to PARGi in PARG KO cells may depend on PARP1/2-dependent pADPr accumulation. To test this hypothesis, CellTiter-Glo and a colony formation assay were used to detect the sensitivity of WT, PARG KO, PARP1/2 DKO, and PARG/PARP1/2 TKO cells to PARGi. As expected, the sensitivity of PARG KO to PARGi was completely reversed in TKO cells, and DKO cells showed moderate resistance compared with WT cells (**Fig. 2C**). Indeed, TKO cells completely rescued cell death induced by PARGi in PARG KO cells (**Fig. S2C**). In addition, treatment with PARPi olaparib was able to reverse PARGi sensitivity in PARG KO cells (**Fig. S2C**). These data together suggest that the cytotoxicity of PARGi in PARG KO cells was due to uncontrolled PARylation, which requires PARP1/2.

PARP1/2 catalyzes the pADPr reaction by transferring the ADR-ribose moiety of NAD^+ to the acceptor proteins with the release of NAM ⁴¹. Uncontrolled PARylation may lead to cytotoxicity, at least in part because of NAD^+ depletion^{23, 42, 43}. To determine whether cell death induced by PARGi in PARG KO cells is at least partially caused by NAD^+ depletion, we measured the relative NAD^+ level in WT and PARG KO cells treated with PARGi and used the nicotinamide phosphoribosyltransferase (NAMPT) inhibitor FK866 as a positive control. FK866 is a noncompetitive inhibition of NAMPT, a key enzyme involved in the regulation of NAD^+ biosynthesis from its natural precursor NAM ⁴⁴. As expected, the NAD^+ level decreased modestly over the treatment period in PARG KO cells, but this decrease was not as dramatic as those observed in cells treated with FK866 (**Fig. S2D**). Moreover, treatment with FK866, which significantly decreased the NAD^+ level, did not lead to cell death (**Fig. S2E**), suggesting that the decreased NAD^+ level is not sufficient or the only reason for the cytotoxicity observed in PARG KO cells treated with PARGi. Furthermore, we treated cells with the NAD^+ precursors nicotinamide mononucleotide (NMN) or NAM ^{23, 45}. As expected, NMN, NAM , and PARPi rescued the NAD^+ decrease in both non-treated and PARGi treated cells (**Fig. 2D**). In addition, NMN and NAM were able to not only reduce pADPr accumulation induced by long-term PARGi treatment (**Fig. 2E**), but also rescue cell lethality caused by PARGi treatment in both HEK293A-derived and HeLa-derived PARG KO cells (**Fig. 2F** and **S2F**). Together, these results suggest that uncontrolled PARylation in the absence of any dePARylation activities would lead to cytotoxicity, probably due to several mechanisms including apoptosis and are not limited to NAD^+ depletion/reduction.

Unexpected S phase pADPr signaling observed in PARG KO cells treated with PARGi

PARP1/2 are activated by DNA damage to generate pADPr signals. Under normal growth conditions, PARP1/2 activities may be low; therefore, it was extremely difficult to detect pADPr. Unexpectedly, pADPr signaling from normal proliferating cells without any exogenous DNA damage was detected in PARG KO cells after PARGi treatment (**Fig. 2B** and **S2B**), which was apparent in S phase cells, as detected by FACS analysis (**Fig. 2B**). To further characterize pADPr and DNA damage signaling induced by PARGi or other DNA damaging agents, we used flow cytometry to detect both pADPr and γH2AX signaling (**Fig. 3A** and **S3A**). Interestingly, pADPr signaling was specifically detected in S phase in PARG KO cells treated with PARGi, while at the same time, DNA damage signaling only showed a slight increase or was not observed, as revealed by anti- γH2AX antibody (**Fig. 3A**). On the other hand, MMS treatment led to increased pADPr signaling throughout the cell cycle (**Fig. 3A**). However, it only led to an S phase-specific increase in DNA damage signaling, i.e. γH2AX . Our interpretation of the MMS results is that MMS treatment would lead to DNA alkylation throughout the cell cycle, which is repaired by the BER pathway. BER creates single-strand nicks as repair intermediates, which are recognized by PARP1 and activate PARP1. Therefore, pADPr signaling can be detected in all cell cycle phases in PARG KO cells under this condition. However, single-strand nicks are not recognized by DNA damage checkpoint

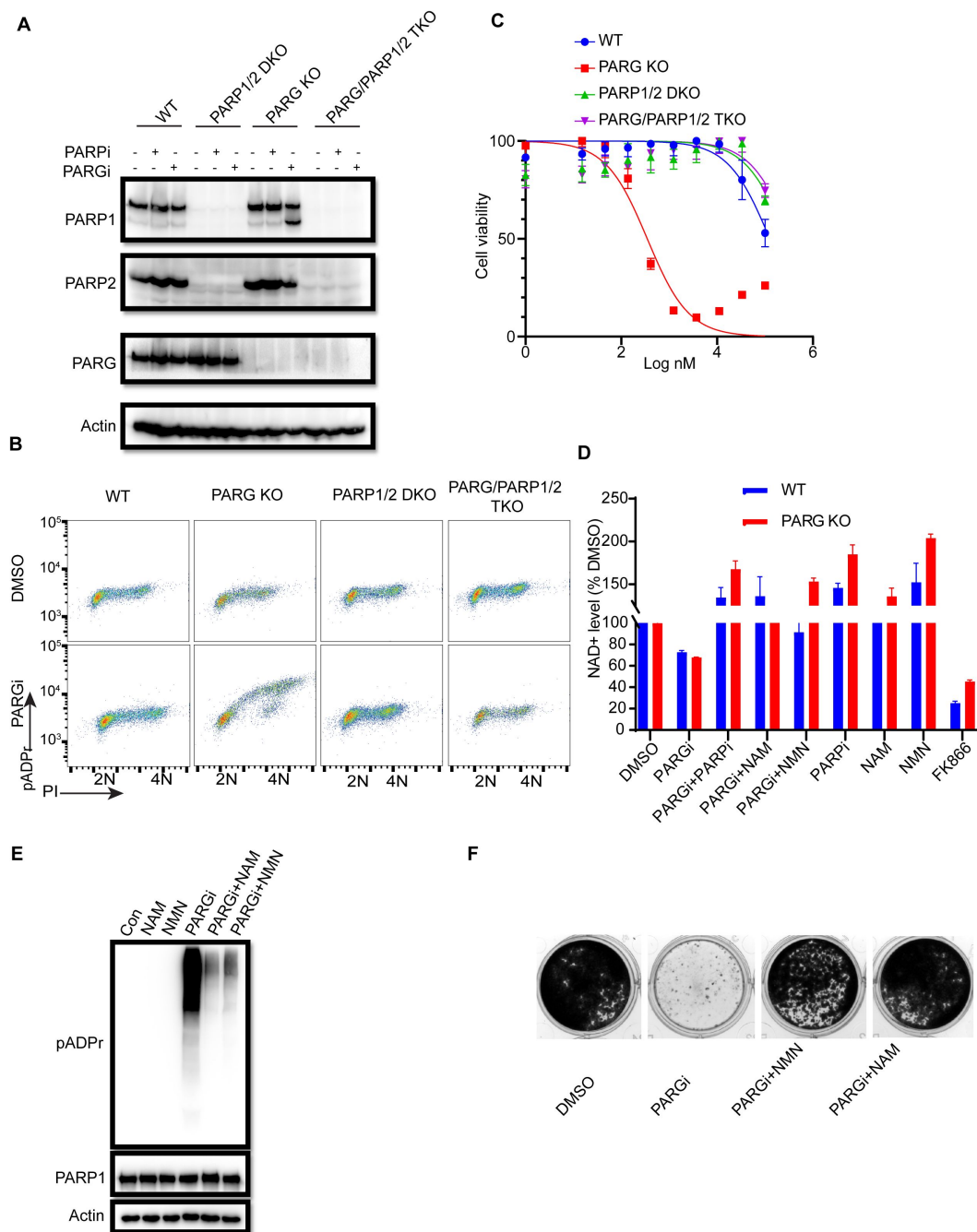


Fig. 2.

PARGi treatment induces NAD⁺- and PARP-dependent cell death in PARG KO cells.

a, HEK293A WT, PARG KO, PARG1/2 DKO, and PARG/PARP1/2 TKO cells were treated with PARGi (1 μ M) for 72 hours. The total cell lysates were immunoblotted with the indicated antibodies. **b**, HEK293A WT, PARG KO, PARG1/2 DKO, and PARG/PARP1/2 TKO cells were treated with DMSO or 10 μ M PARGi for 4 hours and then fixed and stained with anti-pADPr antibody and propidium iodide (PI). **c**, HEK293A WT, PARG KO, PARG1/2 DKO cells, and PARG/PARP1/2 TKO cells were treated with different doses of PARGi for 72 hours. Cell viability was determined by the CellTiter-Glo assay. **d**, Relative NAD⁺ level in HEK293A WT and PARG KO cells with the indicated treatment for 48 hours. PARGi, 10 μ M; PARPi, 10 μ M; NAM, 100 μ M; NMN, 1 mM; FK866, 10 nM. **e**, PARG KO cells were treated with PARGi (10 μ M) or PARGi and NAM (100 μ M) or NMN (1 mM) for 48 hours. The total cell lysates were immunoblotted with the indicated antibodies. **f**, Results of clonogenic assays conducted using HEK293A PARG KO cells treated with PARGi (500 nM) or PARGi and NAM (100 μ M) or NMN (1 mM) for 7 days.

pathways. These nicks, when encountering replication forks, will be converted to DSBs and therefore activate DNA damage checkpoint kinases, resulting in increased γ H2AX signaling specifically in S phase cells.

To further explore the relationship between pADPr and γ H2AX signaling, we conducted a similar flow cytometry analysis in parental WT and PARG KO cells following treatments with different DNA damaging agents. Again, PARGi treatment in PARG KO cells led to an S phase-specific increase in pADPr signaling; this unique pattern was not detected in response to any of the DNA damaging agents used in this study (**Fig. S3A**). Our current working hypothesis is that PARG KO cells treated with PARGi may completely block any dePARylation reactions. The dramatic S phase-specific pADPr signaling detected under this condition indicates that PARP1 is specifically activated in S phase cells.

During DNA replication, maturation of Okazaki fragments involves single-strand nicks on DNA, which is normally ligated by LIG1. These unligated Okazaki fragment intermediates could activate PARP1 to generate endogenous S phase pADPr⁶. It is possible that in such situations, the maturation of Okazaki fragments may require PARP1-dependent recruitment of XRCC1/LIG3⁶, which agrees with the results of early studies that indicate that both LIG1 and LIG3 are required for Okazaki fragment maturation⁴⁶. Of course, PARP1 may have other functions in the S phase, including DNA repair⁴⁷.

The S phase pADPr signaling observed in PARG KO cells after PARGi treatment was observed under conditions in which no or very limited DNA damage or replication stress–induced signaling was detected by anti- γ H2AX antibody (**Fig. 3A**) and S phase pADPr resulting from unligated Okazaki fragments were observed under similar conditions⁶; thus, we reasoned that unligated Okazaki fragments in normal proliferating cells may render this unexpected S phase pADPr signaling in PARG KO cells after PARGi treatment. Indeed, pre-treatment with emetine, which diminishes Okazaki fragments by its anti-protein synthetic activity^{6, 48, 49}, greatly inhibited S phase pADPr signaling in PARG KO cells (**Fig. 3B**). Furthermore, mild inhibition of POLA1 with two inhibitors (i.e. adarotene and CD437) also abolished the PARGi induced S phase pADPr signaling in PARG KO cells (**Fig. S3B**).

To further support S phase-specific pADPr signaling resulting from unligated Okazaki fragments, we performed double thymidine block (DTB) and release experiments. Indeed, release into S phase is critical for the pADPr signaling observed in PARG KO cells treated with PARGi (**Fig. 3C**). Moreover, the cytotoxicity of PARGi also requires S phase progression, since both control proliferating cells and double thymidine blocked and released cells died following PARGi treatment, while cells arrested by DTB failed to do so (**Fig. 3D**). Additionally, mild inhibition of POLA1 partially rescued the cytotoxicity of PARGi in PARG KO cells (**Fig. 3E**).

PARG KO cells show prolonged PARP1 chromatin binding

Following DNA damage, PARP1 detects and is activated by both SSBs and DSBs to initiate subsequent DNA repair^{50, 51}. After that, PARP1 dissociates from DNA. If this does not occur, the persistent PARP1-DNA complexes could trap PARP1 on chromatin, which is one of the key characteristics of PARPi-mediated cytotoxicity³⁵. Given that the sensitivity of PARG KO cells to PARGi resulted from PARP1/2-dependent pADPr accumulation, which indicates an association and activation of PARP1/2 by DNA, we used a previously described trapping assay to measure the levels of chromatin-bound PARP1^{35, 36}. Unexpectedly, the levels of chromatin-bound PARP1/2 increased in PARG KO cells, regardless of treatment, while the soluble fraction of PARP1 decreased compared with that in WT cells (**Fig. S4A**).

To further explore the chromatin-bound PARP1, we performed similar soluble and chromatin fractionation and enrichment of PARylated proteins by Af1521 beads. Consistently, chromatin-bound PARP1 increased in PARG KO cells, regardless of whether these cells were treated with

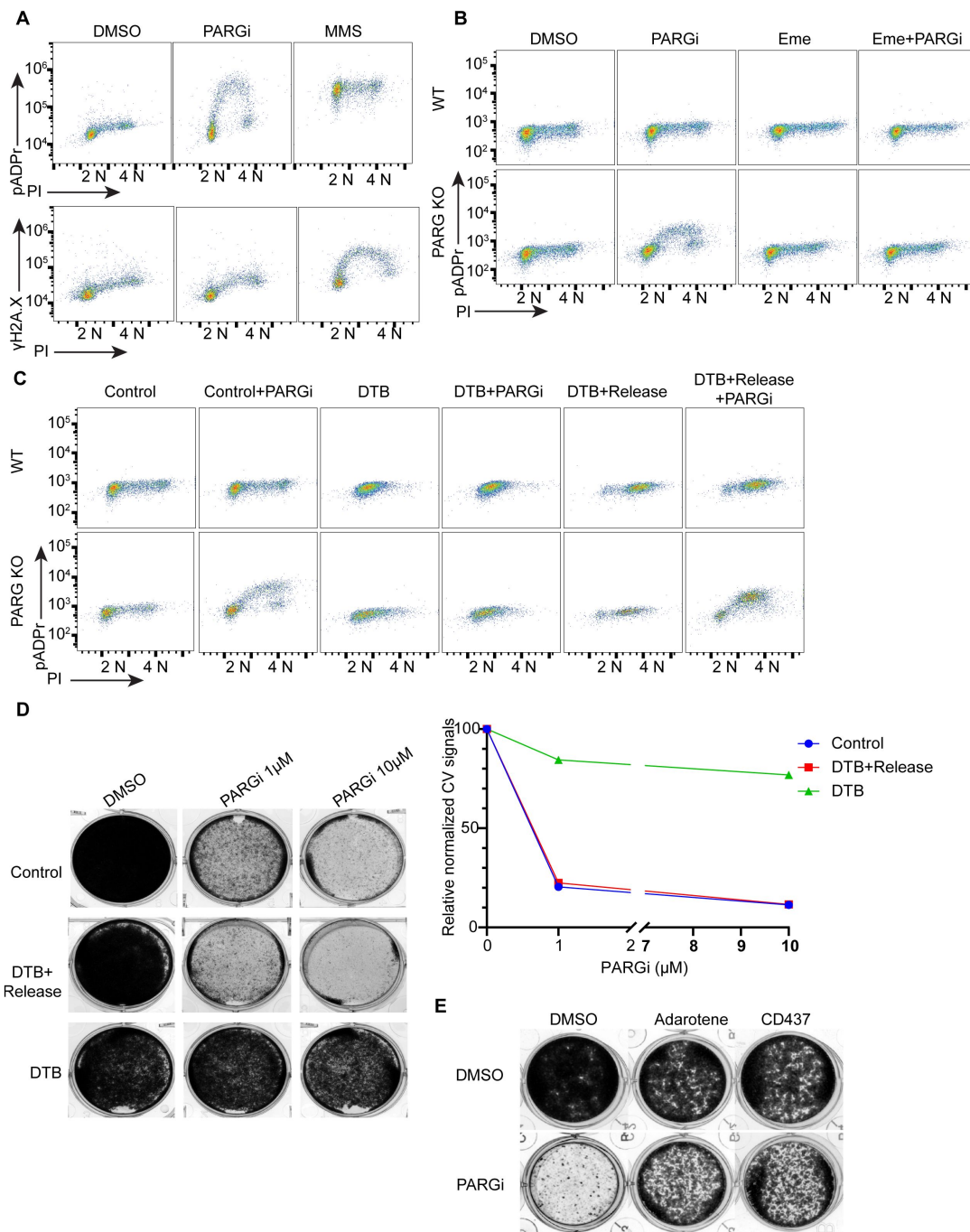


Fig. 3.

PARGi treatment induces S phase-specific pADPr signaling in PARG KO cells.

a, HEK293A WT and PARG KO cells were treated with DMSO or 10 μM PARGi for 4 hours or 0.01% MMS for 30 min and then fixed and stained with anti-pADPr antibody or anti-γH2A.X antibody and PI. **b**, HEK293A WT and HEK293A PARG KO cells were mock-treated or pre-treated with 2 μM emetine for 90 min and then treated with PARGi for an additional 4 hours. Cells were fixed and stained with anti-pADPr antibody and PI. **c**, HEK293A PARG KO cells were synchronized with double thymidine block (DTB). Cells remained with DTB or were released from DTB, treated with 10 μM PARGi for 4 hours, and then fixed and stained with anti-pADPr antibody and PI. **d**, Representative images and results (left) of clonogenic assays conducted using control cells and DTB synchronized or released HEK293A PARG KO cells treated with the indicated doses of PARGi for 7 days, and quantification of crystal violet staining assay (right). **e**, Results of clonogenic assays were conducted in PARG KO cells with indicated treatment for 7 days (PARGi, 1 μM; adarotene, 200 nM; CD437, 800 nM).

PARGi (**Fig. 4A**). Moreover, PARylated PARP1 also increased in chromatin fraction. Interestingly, following PARGi treatment, PARG KO cells showed more pADPr levels than did WT cells in chromatin fraction, while similar pADPr levels were detected in soluble fractions (**Fig. 4A**). Indeed, a recent study by Gogola and colleagues suggest that PARG depletion does not enhance PARP1 dissociation from DNA but prevents excessive PARP1 binding, on the basis of the results of a similar trapping assay and a laser-induced DNA damage assay for measuring PARP1 association⁵². Taken together, these results indicate that prolonged PARP1 chromatin binding or trapping occurs in PARG KO cells, which may contribute to the sensitivity of these cells to PARGi.

To further support the contribution of chromatin-bound PARP1 to PARGi sensitivity, we introduced either WT PARP1 or a trapping-deficient PARP1^{del.p.119K120S} construction into TKO cells (**Fig. S4B**). PARP1^{del.p.119K120S} mutant is a trapping-deficient mutant, which leads to PARPi resistance⁵³.⁵⁴ Notably, expression of WT PARP1 led to dramatic PARGi sensitivity in TKO cells, while PARP1^{del.p.119K120S} mutant only showed a slight increase in PARGi sensitivity (**Fig. 4B**). Together, these data suggest that increased chromatin-bound and trapped PARP1 contributes, at least in part, to the sensitivity of these cells to PARGi.

To investigate the consequence of S phase pADPr and prolonged PARP1 chromatin binding in PARG KO cells, we again used flow cytometry to detect pADPr signaling in a time course following PARGi treatment. Consistent with the aforementioned data, S phase pADPr was detected following short PARGi treatment (i.e. 4 hours), while pADPr signaling was detected throughout the cell cycle following prolonged PARGi treatment, i.e. 24 or 48 hours (**Fig. 4C**). Furthermore, SSBs were not detected following short PARGi treatment, as measured by an alkaline comet assay (**Fig. S4C**), which may due to the sensitivity of this assay. However, prolonged PARGi treatment in PARG KO cells led to dramatic increase of SSBs detected by this assay (**Fig. S4C**). Similarly, we observed S phase-specific pADPr signaling with short PARGi treatment and pADPr signaling throughout the cell cycle with prolonged PARGi treatment in HeLa PARG KO cells, although significant S-phase pADPr was detected in control WT HeLa cells (**Fig. S4D**), which was not observed in control WT 293A cells (**Fig. 4C**). The difference in pADPr signaling between HeLa and 293A cells may due to a low level of PARG and/or high activity of PARP1 in HeLa cells, in which high-molecular weight smears were detected following PARGi and MMS treatment (**Fig. S4E**). Together, these data indicate that PARG regulates S-phase pADPr in normal proliferating cells.

To further confirm pADPr signaling observed by flow cytometry analysis, we detected pADPr and γ H2AX by Western blotting in HEK293A and HeLa cells and their corresponding PARG KO cells, while additional MMS treatment was included as a positive control (**Fig. 4D** and **S4F**). Expectedly, we only observed increased γ H2AX signaling in PARG KO cells with prolonged PARGi treatment (48 hours of treatment in 293A PARG KO cells and 24 and 48 hours of treatment in HeLa PARG KO cells) or those treated with additional MMS, indicating that pADPr accumulation leads to DNA damage throughout the cell cycle. The pADPr signaling in PARG KO cells pre-treated with PARGi was further enhanced in MMS-treated cells (**Fig. 4D** and **S4F**), likely due to increased SSBs; nevertheless, PARGi-mediated pADPr accumulation in PARG KO cells under condition without any exogenous DNA damage was sufficient to cause cytotoxicity. This situation is different from the hyperactivity and progressive inactivation of PARP1 in XRCC1-deficient cells after MMS treatment⁴². Taken together, we speculate that inhibition of dePARylation in PARG KO cells treated with PARGi lead to pADPr accumulation and cell lethality.

CRISPR screens reveal genes responsible for regulating pADPr signaling and/or cell lethality in WT and PARG KO cells

PARG KO and PARGi appear to have a synergic effect on cell viability, which depends on PARP1/2-dependent pADPr accumulation (**Fig. 1** and **2**). Further, we showed that unligated Okazaki fragments were the likely source of SSBs, which leads to S phase pADPr signaling (**Fig. 3**). Moreover, we found that uncontrolled S phase pADPr accumulation eventually leads to DNA

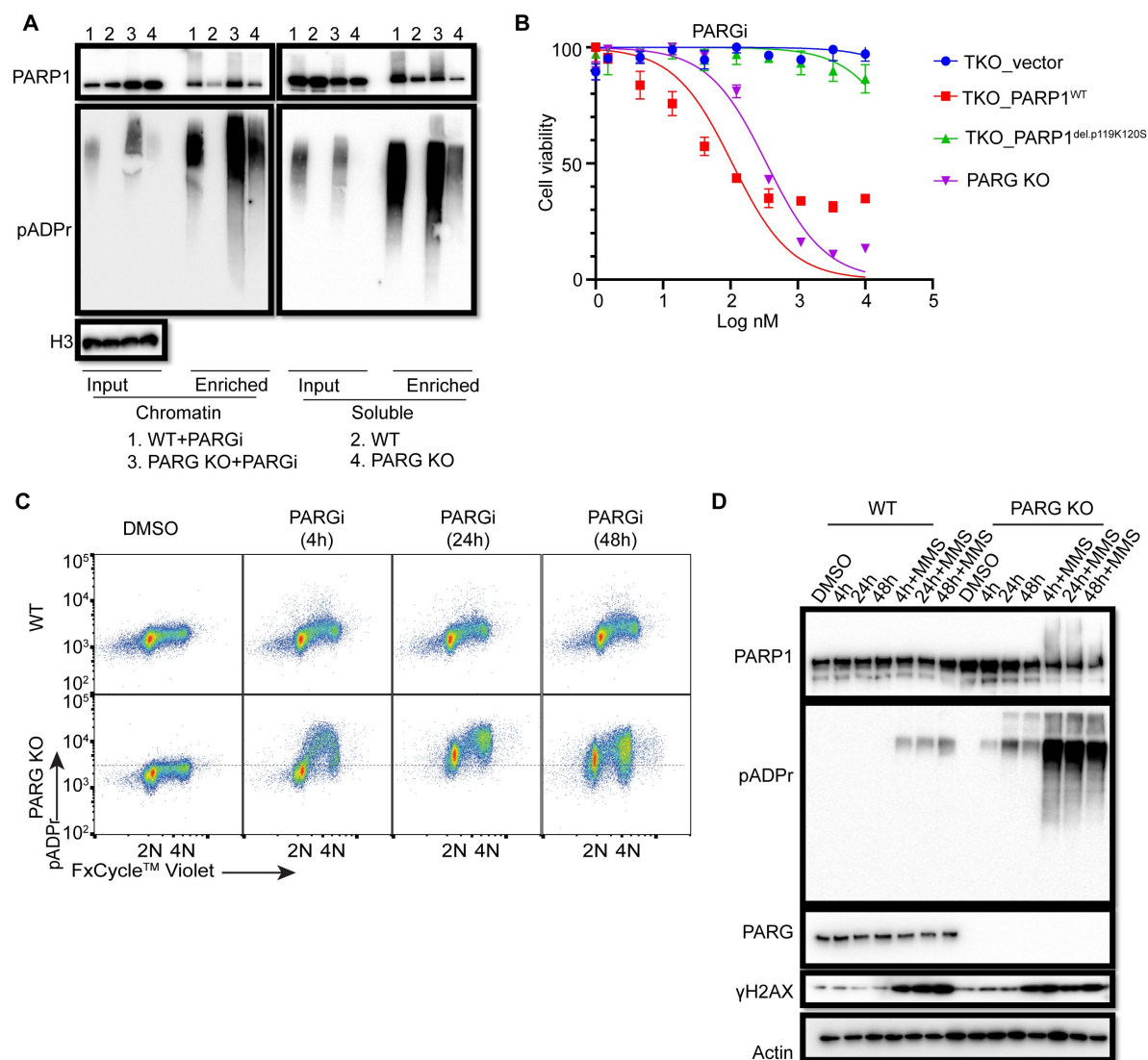


Fig. 4.

Prolonged PARGi treatment induces pADPr throughout the cell cycle and DDR in PARG KO cells.

a, Immunoblots of chromatin-bound PARP1 and PARylated proteins in HEK293A WT and PARG KO cells treated with PARGi (10 μ M) for 4 hours. PARylated proteins were enriched by Af1521 beads. **b**, Sensitivity of HEK 293A PARG/PARP1/2 TKO and PARP1 reconstitution cells to PARGi. Cells were treated with different doses of PARGi for 72 hours, and cell viability was determined by the CellTiter-Glo assay. **c**, Prolonged PARGi treatment induces pADPr throughout the cell cycle in PARG KO cells. HEK293A WT and PARG KO cells were treated with PARGi (10 μ M) for the indicated time and then fixed and stained with anti-pADPr antibody and violet. **d**, Immunoblotting of γ H2A.X signals and other indicated proteins and modifications induced by prolonged PARGi/-MMS treatment.

damage, as revealed by anti- γ H2AX antibody and alkaline comet assay (**Fig. 4**). These data prompted us to further explore potential genes that may be responsible for controlling cell viability and pADPr signaling in control WT and PARG KO cells, with or without PARGi treatment. As we speculated above, there may be a potential second gene targeted by PARGi, which only becomes essential in the absence of PARG. We thus performed the outlined whole-genome CRISPR screens to address these questions (**Fig. 5A**), similar to the studies published by us and others^{37, 55}.

As for FACS-based CRISPR screening, we showed that PARP1 consistently appeared as the gene that, when depleted, led to reduced pADPr in all settings (**Fig. 5B**, **S5A** and **Table S2-4**), which agrees with the dominant role of PARP1 in promoting PARylation in cells. The depleted genes that led to increased pADPr signaling were also similar in all of these settings (**Fig. 5B** and **S5A**); these include BER genes (i.e. POLB, XRCC1, LIG3, and CHD1L) and genes involved in DNA replication/Okazaki fragment maturation (i.e. RFCs, FEN1, and LIG1). Interestingly, ARH3 only appeared in PARG KO cells treated with PARGi (**Fig. 5B** and **S5A**), indicating that as anticipated ARH3 may serve as a backup dePARylating enzyme.

We further investigated whether the aforementioned genes that regulated pADPr signaling would show strong synthetic lethality with PARGi in both PARG KO cells (**Fig. 5C** and **Table S5**) and WT cells (**Fig. 5D** and **Table S6**). To identify the most confident hits, we used a stringent false discovery rate (FDR) threshold of 0.05⁵⁶. Consistently, PARP1 was listed as the most resistant gene with PARGi in PARG KO cells (**Fig. 5C**). Interestingly, ARH3 appeared as the top gene in PARG KO cells treated with PARGi (**Fig. 5C**), while other dePARylation enzymes (e.g. MARCROD1/2 and TARG1/OARD1) did not show any synthetic lethal effect in these screens. These data agree with the results of FACS-based screens and suggest that ARH3 is a backup enzyme in the absence of PARG following PARGi treatment. Indeed, loss of ARH3 further sensitized PARG KO cells to PARGi (**Fig. S5B**). In addition, HPF1, another gene/protein involved in PARylation regulation and licensing serine mono-ADP-ribosylation by PARP1/2⁵⁷, was among the top synthetic lethality genes in PARG KO cells treated with PARGi, but not in WT cells (**Fig. 5C,D** and **S5C**).

Thus, our CRISPR screens revealed that genes involved in PARylation regulation (ARH3 and HPF1), BER (i.e. POLB, XRCC1, LIG3, and CHD1L), and DNA replication/Okazaki fragment maturation (i.e. RFCs, FEN1, and LIG1) are responsible for pADPr signaling and/or cell lethality in WT and PARG KO cells.

To further explore the involvement of these aforementioned genes, we compared synthetic lethality genes in PARG KO cells with those in WT cells treated with PARGi. Interestingly, two BER key genes, POLB and XRCC1, showed strong synthetic lethality with PARGi in both PARG KO cells (**Fig. 5C**) and WT cells (**Fig. 5D**), which is consistent with the results of a previous report that POLB-deficient cells are sensitive to PARGi³⁸. Moreover, the GO analysis revealed that the BER pathway displays synthetic lethality in both PARG KO cells and WT cells following PARGi treatment (**Fig. 5E**). Importantly, LIG1 and LIG3, the two ligases that are directly involved in Okazaki fragment ligation, were also listed as synthetic lethality genes (**Fig. 5C** and **5D**). To further validate these data, we created several KO/KD cell lines, including KO of BER genes (i.e. POLB and XRCC1) and genes involved in Okazaki fragment ligation (i.e. LIG1 and LIG3) (**Fig. S5D**). Consistent with previous reports, XRCC1 loss destabilizes LIG3 (**Fig. S5D**)^{58, 59}. Notably, loss of genes involved in Okazaki fragment ligation induced S phase pADPr signaling, just like PARG KO cells following PARGi treatment, while loss of BER genes resulted in pADPr signaling throughout the cell cycle (**Fig. S5E**). These results agree with the aforementioned results that unligated Okazaki fragments are the likely source of SSBs, which induces S phase-specific pADPr signaling. Also consistent with CRISPR screening data, loss of genes involved in Okazaki fragment ligation or BER resulted in increased sensitivity to PARGi (**Fig. 5E**). Furthermore, we knocked down POLB in WT and PARG KO cells (**Fig. S5F**), which resulted in increased sensitivity to PARGi in both WT and

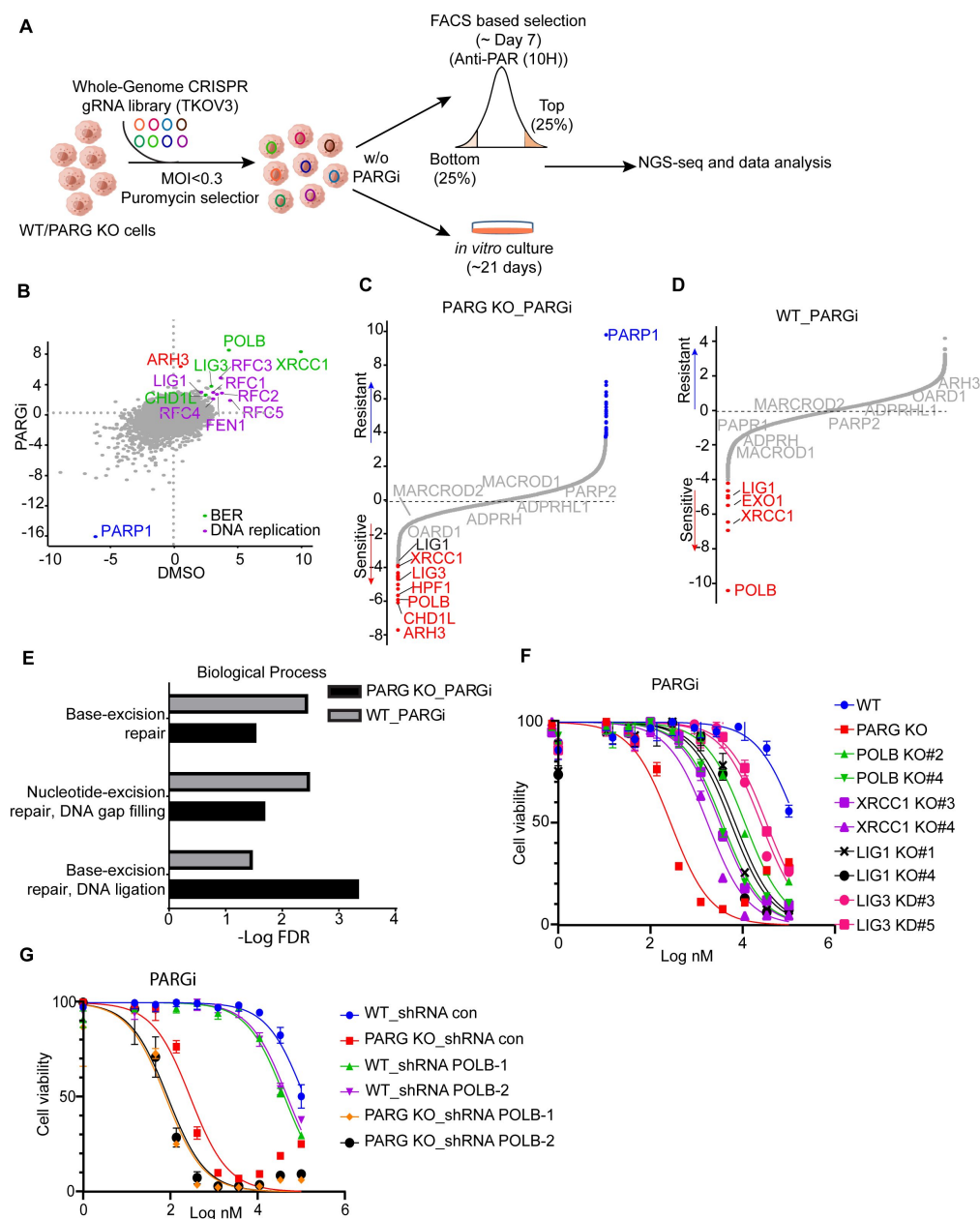


Fig. 5.

CRISPR screening identifies regulators of pADPr and cell viability.

a, Workflow of whole-genome CRISPR screens. For FACS-based CRISPR screening, 5 days after puromycin selection, cells were treated with PARGi (10uM) for 4 hours and then stained with anti-pADPr antibody and sorted with flow cytometry. Cells with strong signals (top 25%, TOP) and weak signals (bottom 25%, BOT) were selected. The sgRNAs from these cells were then sequenced and analyzed. For cell viability screening, cells were treated with or without PARGi for 21 days before collection. **b**, Scatter plot of DrugZ scores of PARG KO cells treated with or without PARGi treatment. The genes in the same pathway were marked with specific colors. A positive score indicates an enhanced pADPr signal, while a minus score indicates a decreased pADPr signal. **c** and **d**, Ranking of PARGi co-essential genes on the basis of a DrugZ analysis of the results of CRISPR/Cas9 screens performed with Toronto Knock Out Library (version 3) in HEK293A PARG KO cells and HEK293A cells. **e**, Analysis of biological processes of PARGi co-essential genes identified in HEK293A PARG KO cells and HEK293A cells. **f**, HEK293A WT cells, PARG KO, POLB KO, LIG1 KO, XRCC1 KO, and LIG3 knockdown cells were treated with different doses of PARGi for 72 hours. Cell viability was determined by the CellTiter-Glo assay. **g**, The cell Viability of HEK293A WT and PARG cells under POLB knockdown to PARGi. Cells were treated with different doses of PARGi for 72 hours.

PARG KO cells (**Fig. 5G**). Together, our data indicate that unligated Okazaki fragments induce S phase-specific pADPr, while BER intermediates lead to pADPr throughout the cell cycle; deficiency in either of these events would increase cytotoxicity following PARGi treatment.

PARG expression is a potential biomarker for PARGi-induced cytotoxicity

We showed that cells with no detectable full-length PARG expression are extremely sensitive to PARG inhibition (**Fig. 1** and **S1**). Thus, we are interested in determining whether PARG expression would correlate with PARGi sensitivity. An early study examined a panel of ovarian cancer cell lines and reported PARGi sensitivity in a subset of these cell lines³⁹. We compared PARG expression in PARGi sensitive and resistant cell lines and found that PARG expression was significantly reduced in PARGi sensitive cell lines (**Fig. 6A**). To support this result, we knocked down PARG by shRNA in HeLa cells and showed that loss or further reduction of PARG led to dramatic sensitivity to PARGi (**Fig. 6B**). To further test our hypothesis, we ranked the PARG mRNA level in several ovarian cancer cell lines using the CCLE database (**Fig. 6C**). We identified an ovarian cancer cell line, RMUGS, which has low PARG expression, similar to that observed in another known PARGi sensitive cell line, KURAMOCHI. Notably, RMUGS cells were considered as potential PARGi-resistant cells based on an earlier report³⁹. However, we showed that RMUGS cells were very sensitive to PARGi treatment (**Fig. 6D**), and the sensitivity of these cells to PARGi could be reversed by PARPi treatment (**Fig. 6E**). These data indicate that PARG expression may dictate the sensitivity of tumor cells to PARGi-based therapy.

An early study reported that loss of PARG expression resulted in PARPi resistance⁵². They demonstrated that PARG is frequently lost in acquired PARPi-resistant mouse mammary tumors and further revealed that PARG depletion occurs in triple-negative breast and ovarian cancer. Another study showed that 60 of 274 (22%) of human ovarian tumors have low PARG expression³⁸. Thus, PARGi can be potentially used to treat these PARPi-resistant cancers, especially those with low PARG expression. We further performed an IHC assay to detect PARG expression in breast and ovarian cancer TMA samples. The IHC assay was established with the use of our xenograft tumors derived from HeLa and HeLa PARG KO cells (data not shown). As shown in **Figure 6F**, PARG expression was detected in normal ovarian and breast tissues. PARG expression was also detected in a majority or most of ovarian and breast tumor samples (**Fig. 6F**). Interestingly, about 5% to 40% of these breast or ovarian cancer tumor samples showed low expression levels of PARG, some even showed no detectable level of PARG, compared with those in normal tissue samples (**Fig. 6F**). These data suggest that PARG downregulation occurs in breast and ovarian cancers. PARGi may be a promising strategy for the treatment of these cancers with low PARG expression.

In early studies, HR-deficient cells showed increased sensitivity to PARG inhibition^{18, 25, 26, 60}. Similarly, we observed PARGi-induced cytotoxicity in RPE1 Flag-Cas9 TP53/BRCA1 DKO cells, but not in control cells (**Fig. S6A**). Moreover, combining PARPi and PARGi did not reveal any additive effect (**Fig. S6A**). Furthermore, we created inducible BRCA1 depletion cells using the auxin-inducible degron (mAID) tag. As shown in **Figure S6B**, HR-proficient cells showed normal sensitivity to PARGi, while HR deficiency by the loss of BRCA1 conferred modest sensitivity to PARGi. The PARGi sensitivity in BRCA1 depletion cells was reversed to normal by the restoration of HR due to 53BP1 loss. Consistent with the results obtained in RPE1 cells, the PARPi+PARGi combination did not lead to any further change in cytotoxicity. While these results indicate that PARGi can be used to target HR-deficient cancers, the effect is quite modest. Moreover, PARGi treatment did not show any effect on PARPi-resistant cells due to restoration of HR mediated by 53BP1 loss.

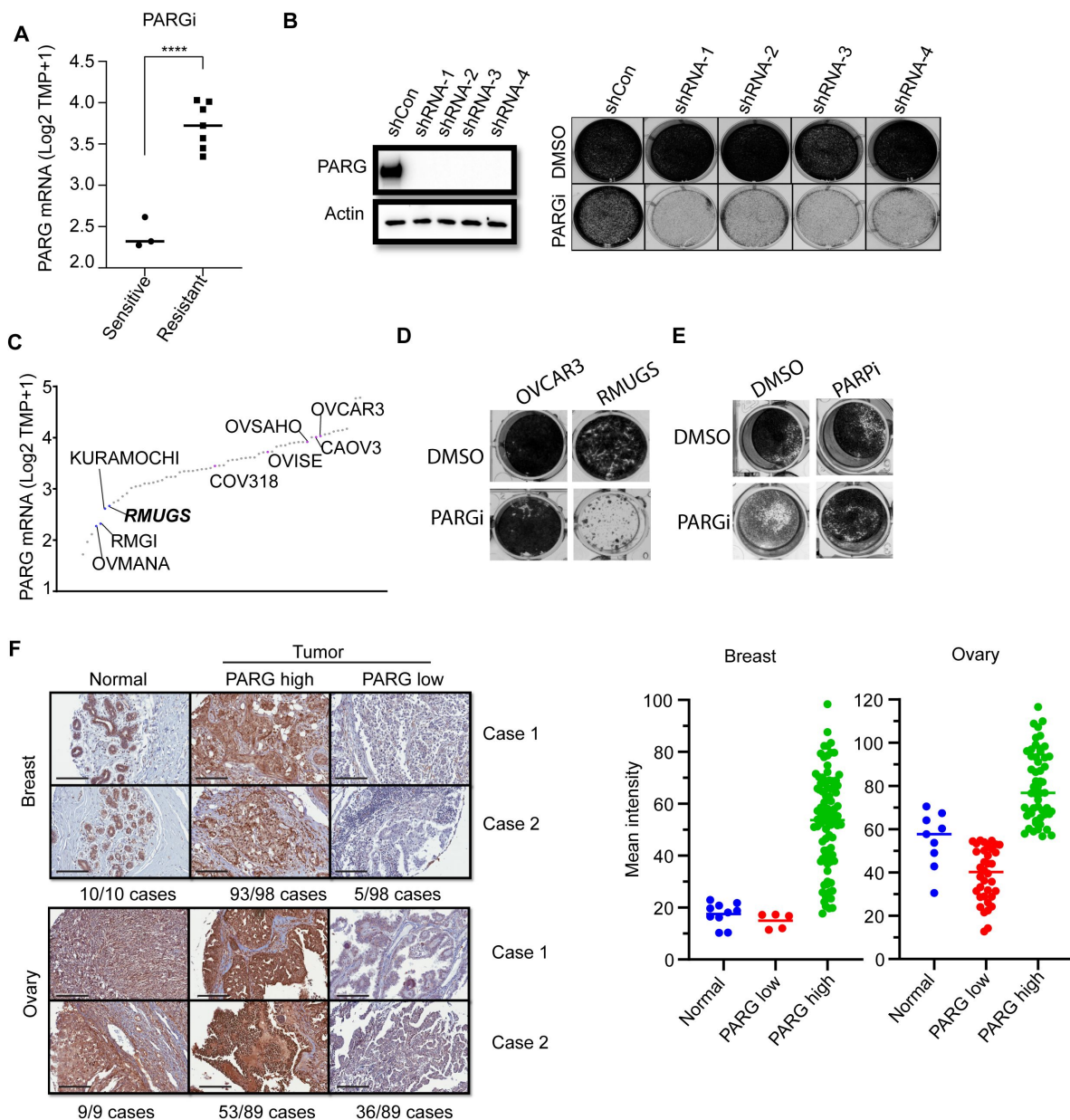


Figure 6.

PARG expression is a potential marker for PARGi sensitivity.

a, PARG mRNA level comparison between sensitive (RMGI, KURAMOCHI, and OVMANA) and resistant cells (COV362, COV318, OV56, OVISE, OVSAHQ, CAOV3, and OVCAR3) from Pillay's work on the basis of CCLE data. **b**, Clonogenic assay results of control and PARG knockdown HeLa cells with PARGi (2 μ M) treatment for 7 days. shRNA knockdown efficiency was confirmed by an immunoblot of PARG. **c**, Ranked PARG expression level in ovarian cancer cell lines based on the CCLE database. The sensitive and resistant cells from Pillay's work were labeled. **d**, Clonogenic assay results of OVCA3 and RMUGS treated with or without PARGi (2 μ M). **e**, Clonogenic assay results of RMUGS treated with PARGi (2 μ M), PARPi (2 μ M), or both. **f**, Left: Representative images of PARG IHC staining in breast and ovarian tissues and tumor samples to determine the PARG expression level. The summary is listed at the bottom. Scale bar, 200 μ m. Right: The scatter plot of mean immunostaining intensity of PARG in each sample. The mean of each group was plotted.

To further investigate the potential contribution of HR deficiency and/or PARG loss to PARGi sensitivity, we knocked down PARG by shRNA in these inducible BRCA1 depletion cells with or without 53BP1 KO (**Fig. S7A**). As expected, PARG loss is the main driver of PARGi sensitivity; regardless whether these cells are HR-proficient, HR-deficient, or HR-restored because of 53BP1 loss (**Fig. S7B**). Our data therefore strongly suggest that PARG expression is the major determinant of PARGi sensitivity.

PARG is essential for cell survival

Our results suggested that cells with no detectable full-length PARG expression or low expression of PARG are extremely sensitive to PARG inhibition (**Fig. 1** and **6**). As aforementioned, the question is why PARG KO cells would be sensitive to PARGi. To address this question, we first performed whole-genome CRISPR screening. However, the potential second target was not revealed by our CRISPR screening data (**Fig. 5**). The second possibility is that our PARG KO cells are not complete PARG KO cells. These cells may have residual PARG expression or activity and only cells with very low PARG expression are sensitive to PARGi, as shown in **Fig. 6**. Indeed, two independent Parg knockout mice have been created by targeting early exons of PARG^{13, 61}. While one group of Parg knockout mice showed embryonic lethality and cells derived from these mice only survived in the presence of PARPi¹³, the other Parg knockout mice did not display any major phenotypes, probably because truncated Parg and Parg activity were still detected⁶¹. We favor the second possibility, especially since the PARG catalytic domain is at its C-terminus.

To test this possibility, we first used two independent antibodies that recognize the C-terminus of PARG to detect potential residual PARG isoforms or spliced variants (**Fig. S9A**). Unfortunately, besides full-length PARG, these antibodies also recognized several other bands, some of them were reduced or absent in PARG KO cells, others were not. Thus, we could not draw a clear conclusion which functional isoform/truncated form was expressed in our PARG KO cells. Then, we used two additional gRNAs which target respectively the beginning of the catalytic domain and the sequence around the catalytic site (**Fig. 7A**). Interestingly, we were only able to obtain viable clones with no detectable PARG in the presence of PARPi in either 293A or HeLa cells, which we named as PARG complete/conditional KO cells (cKO) (**Fig. 7B**). Indeed, these cKO cells could not survive without PARPi (**Fig. 7C**). To further investigate whether cell lethality is associated with PARG activity in these cells, we reconstituted PARG cKO cells with WT PARG or a PARG catalytic-inactive mutant. Consistent with previous result (**Fig. 1** and **S1**), WT PARG not only reduced pADPr level but also rescued cell lethality due to PARG loss, while the catalytic-inactive mutant PARG failed to do so (**Fig. 7D**).

The data above indicate that our PARG KO cells have residual PARG activity and treatment with PARGi in these cells further decreases PARG activity, which mimic the complete loss of PARG as that in PARG cKO cells. To further test this hypothesis, we measured the relative PARG dePARylation activity in whole cell lysates (WCL) prepared from different cell lines (**Fig. S9C**). Consistent with our working hypothesis, residual PARG activity was detected in WCL prepared from PARG KO cells, while PARG activity was barely detected or absent in WCL isolated from PARG cKO cells. Please note that we incubated cell lysates with substrates overnight to evaluate the maximum level of pADPr hydrolysis, i.e. PARG activity, we were able to detect in these assays. It is very likely that the PARG activity in PARG KO cells was much lower than that indicated in **Fig. S9C**, due to saturation of signals for lysates isolated from wild-type cells. Thus, the data presented here may under-estimate the reduction of PARG activity in PARG KO cells. Nevertheless, these data indicate the residual PARG activity in PARG KO cells, which is absent in PARG cKO cells. Furthermore, the PARG activity was further inhibited by PARGi in a dose-dependent manner in WCLs prepared from both WT and PARG KO cells (**Fig. S9D-E**). Notably, the PARG activity was consistently lower in WCL prepared from PARG KO cells when compared with that in control wild-type cells in the presence of a wide range of PARGi concentrations. However, a significant fraction of PARG activity still existed in WCL prepared from wild-type cells even at the highest

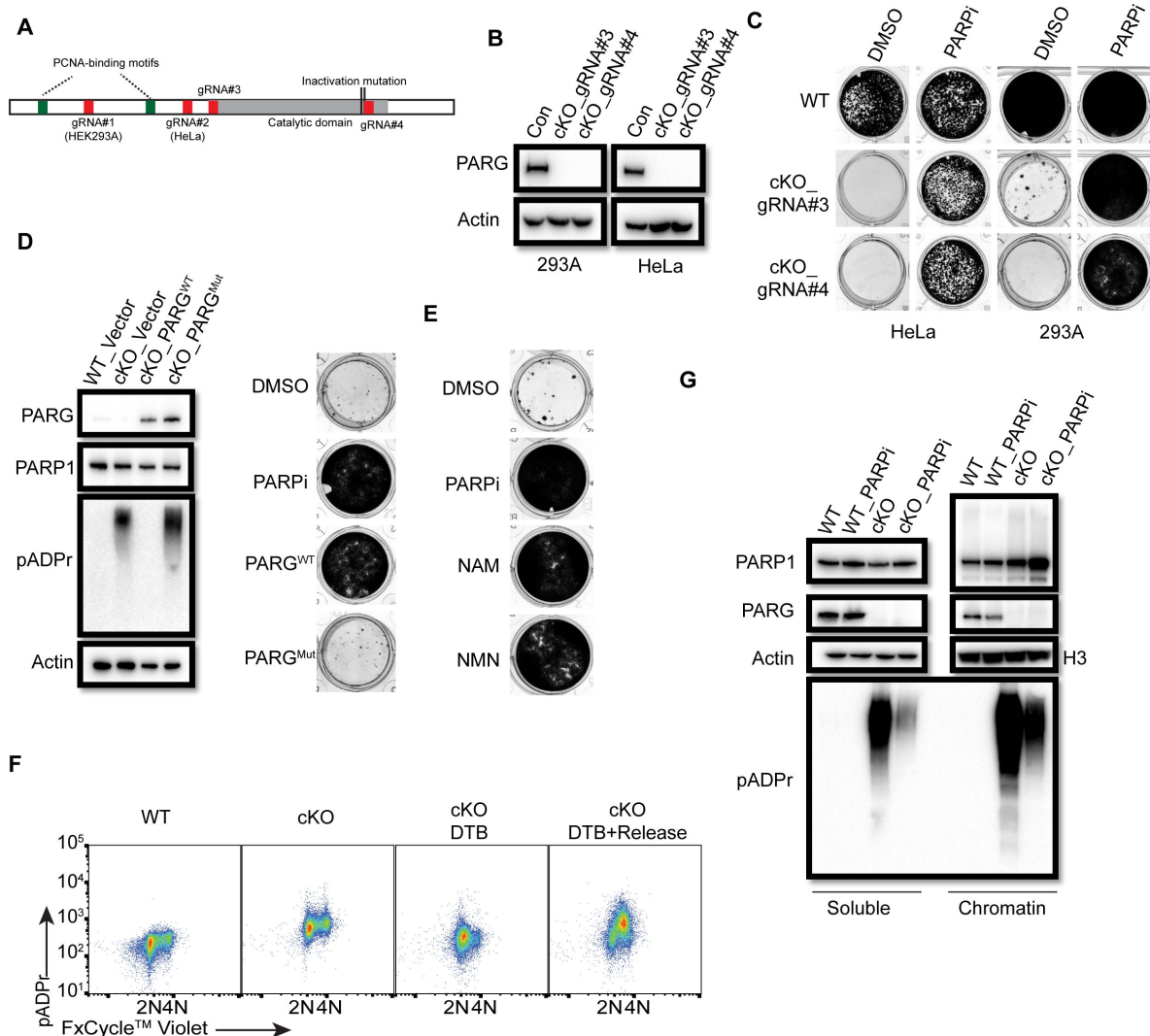


Figure 7.

PARG is essential for cell survival.

a, Diagram of full-length PARG was presented with the indicated gRNAs (gRNA#3 and 4) which target different regions in the C-terminal catalytic domain. The boundary of the catalytic domain was depicted based on Uniprot annotation. gRNA#1 and gRNA#2 were used previously to generate the aforementioned HEK293A- and HeLa-derived PARG KO cells, respectively; while gRNA#3 and gRNA#4 were used to generate PARG complete/conditional knockout (cKO) in the presence of olaparib in HEK293A and HeLa cells. **b**, Immunoblotting was conducted to confirm the loss of PARG in PARG cKO cells derived from HEK293A and HeLa cells, which were cultured in the presence of 100 nM olaparib. **c**, Clonogenic assay results of WT and PARG cKO cells treated with or without PARPi (100nM) for 7 days. **d**, Left: The immunoblots to confirm reconstitution with WT PARG or catalytic inactivation PARG in HEK293A PARG cKO cells. Right: Results of clonogenic survival assay with HEK293A PARG cKO cells reconstituted with WT or catalytic inactivation mutant of PARG for 7 days. **e**, Representative clonogenic results conducted in HEK293A PARG cKO cells treated with NAM (100 μM) or NMN (1 mM) for 7 days. **f**, HEK293A PARG cKO cells were synchronized with double thymidine block (DTB). Cells remained with DTB or were released from DTB for 4 hours, and then fixed and stained with anti-pADPr antibody and FxCycle Violet dye. **g**, Immunoblots of soluble and chromatin-bound PARP1 and pADPr levels in HEK293A WT and PARG cKO cells treated with DMSO or olaparib (10 μM) for 2 hours.

concentration of PARGi used in this study (Fig. S9E). These data indicate residual PARG activity in PARG KO cells likely accounts for the survival of these cells. Moreover, further inhibition of this activity by PARGi likely leads to cell lethality.

We showed above that NAD⁺ precursors (NAM and NMN) were able to rescue cytotoxicity of PARGi in PARG KO cells, probably by NAD⁺ depletion and also that these precursors can also inhibit PARP1/2 activities. Consistently, PARG cKO cells could survive in the presence of NAD⁺ precursors (NAM and NMN) (Fig. 7E). Taken together, PARG is an essential gene and our PARG KO cells treated with PARGi can mimic the complete loss of PARG activity. These data agree with the genetically engineered mouse models as we discussed above¹³.

We showed that S phase pADPr signals were specifically detected in PARG KO cells treated with PARGi, which likely originated from unligated Okazaki fragments (Fig. 3). As expected, we observed S phase-specific pADPr in PARG cKO cells under double-thymidine block and release condition (Fig. 7F). Moreover, the levels of chromatin-bound PARP1 and pADPr signal, which reflects PARylated PARP1 and/or other PARP1 substrates, increased in PARG cKO cells (Fig. 7G). Treatment with PARPi decreased pADPr signals but increased chromatin-bound PARP1 (Fig. 7G), which is consistent with the results obtained in PARG KO cells (Fig. 4A). Together, PARG, as an essential gene, suppresses initial S phase pADPr signaling and PARP1 chromatin accumulation, which eventually lead to cell death caused by uncontrolled pADPr accumulation throughout cell cycle.

Discussion

Aberrant accumulation of pADPr and/or NAD⁺ exhaustion disrupts multiple cell processes, including DNA repair, replication stress response, and transcription regulation; therefore, it is highly toxic and eventually leads to cell death^{10, 22, 42, 43, 62}. Although ADP-ribose hydrolases, especially PARG, have a protective effect against excessive PARP1 engagement, it remains unclear when and how PARG activity is engaged during normal cell proliferation. In this study, we showed that the major function of PARG is to regulate S phase-specific pADPr by PARP1, which is activated by unligated Okazaki fragments. When PARG activity is inhibited, these uncontrolled pADPr would eventually result in DNA damage and cell death. Using unbiased FACS- and cell viability-based genome-wide CRISPR screens, we uncovered additional pADPr modulators, which include the genes involved in BER (POLB, XRCC1, and LIG3), Okazaki maturation (LIG1 and FEN1), ARH3, and HPF1. More importantly, we determined that PARG expression is a critical biomarker for PARGi sensitivity. Furthermore, we showed that PARG is an essential gene by suppressing pADPr accumulation. Taken together, our study uncovers a normal function of PARG in proliferating cells: it removes pADPr generated by PARP1 at unligated Okazaki fragments during DNA replication. Of course, this normal function of PARG is also required under conditions of excessive PARP1 engagement because of defects in Okazaki fragment maturation and/or BER.

PARylation is a reversible post-translational modification. In this study, we showed that timely and efficient dePARylation is critical for cell survival. Thus, both PARylation and dePARylation play important functions in the cell. Particularly, dePARylation appears to be indispensable, since it is required for cell survival. The successful clinical application of PARylation inhibition, such as PARPi for the treatment of BRCA-deficient cancers, raises the possibility of targeting dePARylation as an alternative strategy, especially for the treatment of PARPi-resistant tumors. Given that PARG accounts for more than 90% of dePARylation activity^{7, 29}, selective PARG inhibitors are being developed as potential anti-cancer agents. Although defective HR and DDR have been shown to be synthetic lethal with PARG inhibition, comprehensive investigations of cellular response to PARG inhibition have not yet been conducted. In this study, we showed that unexpectedly, PARG is the top synthetic lethality gene in the CRISPR screening with our homemade DDR sgRNA library, which we further validated in HEK293A and HeLa PARG KO cells (Fig. 1B and S1D). Our PARG

KO cells plus PARGi mimic complete loss of PARG activity (**Fig. S9**), which provides is an unique opportunity to study the major function of PARG in proliferating cells. These data were further confirmed with the use of PARG cKO cells we generated (**Fig. 7**).

PARP1 plays important roles during normal S phase, especially at DNA replication forks and two-ended DSBs⁴⁷. More recently, unligated Okazaki fragments were identified as the source of SSBs that activate PARP1 in normal proliferating cells⁶. Moreover, unligated Okazaki fragments, on which PARP1 may be trapped, can be converted into structures that result in synthetic lethality in HR-defective cancer cells⁶³. PARG is also enriched at DNA replication forks through pADPr- and PCNA-dependent mechanisms^{14, 64, 65}. In addition, PARG may protect DNA replication forks during normal and damage-treated S-phase^{15, 31, 66}. Interestingly, the identification of unligated Okazaki fragments as the source of endogenous pADPr in normal proliferating cells was only noticeable under short-term PARG inhibition. Treatment with another recently developed PARG inhibitor COH34, also led to increased PARylation at replication sites in S phase cells²⁶. These previous reports agree with the data presented in this study. Indeed, we showed robust and specific PARP1/2-dependent S phase pADPr in PARG KO cells treated with PARGi (**Fig. 3A** and **S3**). We further confirmed that the S phase-specific pADPr requires normal S phase progression and the generation of Okazaki fragments (**Fig. 3B-C** and **S3B**). Moreover, the cytotoxicity of PARGi requires S phase progression (**Fig. 3D-E**). In addition, PARG KO cells displayed more chromatin-bound PARP1 and PARylated PARP1 (**Fig. 4A** and **S4A**). Enhanced pADPr levels were observed in the chromatin fraction of PARG KO cells under PARGi treatment (**Fig. 4A**). One possible explanation is that in PARG KO cells treated with PARGi, chromatin-bound or trapped PARP1 may not be removed appropriately or efficiently, which further prevents Okazaki fragment ligation/maturation that eventually results in cell lethality. However, the underlying mechanism of more chromatin-bound PARP1 and PARylated PARP1 in PARG KO cells still needs further elucidation.

The inability to remove S phase-specific pADPr eventually led to pADPr throughout cell cycle and γ H2AX signaling (**Fig. 4C-D** and **S4D-F**), indicating that it is essential to resolve S phase-specific pADPr to maintain cell viability. In agreement with the literature^{6, 26}, we observed S phase-specific pADPr in HeLa cells following PARGi treatment, although such S phase-specific pADPr was not detected in HEK293A cells following PARGi treatment. The lower level of PARG expression may be the reason for this difference between HEK293A and HeLa cells (**Fig. S4E**). However, significant amounts of PARG activity still existed in HeLa cells, since these cells did not show noticeable PARGi sensitivity unless we knockout or knockdown PARG.

Cell viability-based genome-wide CRISPR/Cas9 screening unbiasedly uncovers sensitivity and resistance genes under certain conditions³⁶, while FACS-based CRISPR/Cas9 screening using reporter cell lines or antibodies recognizing specific signaling molecules can reveal key regulators of signaling pathways³⁷. In this study, we performed both types of screens, which revealed key pADPr regulators, including BER pathway proteins (POLB, XRCC1, LIG3, and CHD1L), DNA replication/Okazaki fragment maturation proteins (RFCs, FEN1, and LIG1), and ARH3 (**Fig. 5B** and **S5A**). In addition, consistent with our hypothesis that pADPr accumulation leads to cell death, we showed that loss of genes involved in Okazaki fragment maturation or BER resulted in PARGi sensitivity (**Fig. 5C-G**). All of these results further validated our working model that S phase-specific pADPr likely originated from unligated Okazaki fragments, which if not removed would eventually lead to DNA damage and cell death.

During DNA replication, Okazaki fragments are normally ligated by LIG1. However, some unligated Okazaki fragments may activate PARP1 and recruit XRCC1/LIG3⁶, which agrees with the results of early studies indicating that both LIG1 and LIG3 are required for Okazaki fragment maturation⁴⁶. Indeed, we showed that LIG1 depletion, which would result in more DNA nicks in S phase cells, led to the enhancement of S phase PARylation (**Fig. S5E**). As for LIG3, it can substitute LIG1 function for the ligation of Okazaki fragments^{46, 67}. However, it also has a critical and

essential function in mitochondria but is separated for its role in XRCC1-dependent SSB⁶⁸. We observed minor S phase pADPr due to LIG3 knockdown with PARGi treatment (**Fig. S5E**), which is likely due to insufficient KD of LIG3 in these experiments. Unlike LIG1 KO and LIG3 KD cells, XRCC1 KO and POLB KO cells increase PARylation in a cell cycle-independent manner (**Fig. S5E**), which agrees with the finding that POLB and XRCC1 are required for BER/SSB repair, but not specifically for Okazaki fragment ligation/maturation. Our results suggest that PARylation and dePARylation are mainly involved in two cellular processes, i.e. Okazaki fragment maturation and BER/SSB repair, since both processes have nicked DNA as intermediates, which are likely the physiological substrates that recruit and activate PARP1/2.

Interestingly, two well-known PARylation regulators, ARH3 and HPF1, showed synthetic lethality with PARGi in PARG KO cells (**Fig. 5C**). Given that ARH3 and PARG are the primary dePARylation enzymes¹⁰, it should be predictable that ARH3 was listed as the top hit with PARGi in PARG KO cells in both pADPr signal screen and synthetic lethality screen (**Fig. 5B** and **C**). As for HPF1, a previous report showed that the loss of HPF1 will release the activity of PARP1 to PARylation in acidic residues⁵⁷; moreover, a recent study shows that HPF1 and nucleosomes mediate a dramatic switch in the activity of PARP1 from polymerase to hydrolase⁶⁹. More important, HPF1 promotes the ligation of the Okazaki fragment by LIG3-XRCC1 as a backup pathway⁷⁰, while the unligated Okazaki fragments were the source of the S phase pADPr signal in this study.

More importantly, our data indicate that PARG expression is a potential biomarker for PARGi sensitivity. We used the available database and cell lines to validate that cells with low PARG expression are sensitive to PARGi. In addition, we examined PARG expression by the IHC assay in commercial TMA samples and uncovered a fraction of breast and ovarian cancers with no detectable or low PARG expression (**Fig. 6**). We also examined the sensitivity of HR-deficient cells to PARGi. Although HR deficiency sensitizes cells to PARGi, our results indicate that PARG loss or low expression is still the main driver of PARGi sensitivity in these cells (**Fig. S6,7**).

Mechanically, we showed that the dePARylation activity of PARG is indispensable for cell survival (**Fig. 7** and **S9**). The residual PARG dePARylation activity observed in PARG KO cells likely supports cell growth, which can be further inhibited by PARGi (**Fig. S9**). Although a dose-dependent inhibition of PARG activity by PARGi was also noted in wild-type cells, significant fraction of PARG activity remained even in the presence of highest concentration of PARGi used in this study, which is consistent with our results that HEK293A cells are insensitive to PARGi. More importantly, these data further indicate that PARG expression/activity is a potential biomarker for PARGi sensitivity.

Taken together, the dramatic S phase-specific pADPr signaling detected under our experimental conditions provides us a rare opportunity to gain a better understanding of PARG function and pADPr signaling in normal proliferating cells without any exogenous DNA damage. In addition, our systematic CRISPR/Cas9 screens uncover key regulators of pADPr signaling that also contribute to PARGi sensitivity. Moreover, our finding that PARG expression may a potential biomarker of PARGi sensitivity will allow the further development of these inhibitors for the treatment of tumors with PARG loss and therefore offer a new targeted strategy for cancer therapy.

Competing interest statement

The authors declare that they have no conflicts of interest with the content of this article.

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Author contributions

L.N., C.W., B.G., L.M., and J.C. conceived the project; L.N. performed most of the experiments and analyzed the data with assistance from C.W., X.L., M.H., S.L., X.F., H.T., Q.H., and X.S.; C.W. performed all CRISPR/CAS9 screenings and the data analysis; M.T. provided inducible BRCA1 depletion cells using the auxin-inducible degron (mAID) tag; L.N. and J.C. wrote the manuscript with input from all authors; and all authors commented on the manuscript.

Data availability

All data needed to evaluate the conclusions herein are presented in the article or Supplemental Information. All of the knockout and complete knockout cell lines generated in this study are validated by DNA sequencing (**Fig. S8** and **S9B**), which can be requested with the completion of a material transfer agreement. The inducible BRCA1 depletion cells with the use of auxin-inducible degron (mAID) tag cell lines can be provided by J.C. pending scientific review and completion of a material transfer agreement. Requests for these cell lines should be submitted to J.C. via e-mail at JChen8@mdanderson.org.

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Editors

Reviewing Editor

Tony Hunter

Salk Institute for Biological Studies, La Jolla, United States of America

Senior Editor

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Icahn School of Medicine at Mount Sinai, New York, United States of America

Reviewer #2 (Public Review):

Summary:

In this manuscript Nie et al investigate the effect of PARG KO and PARG inhibition (PARGi) on pADPR, DNA damage, cell viability and synthetic lethal interactions in HEK293A and Hela cells. Surprisingly, the authors report that PARG KO cells are sensitive to PARGi and show higher pADPR levels than PARG KO cells, which is abrogated upon deletion or inhibition of PARP1/PARP2. The authors explain the sensitivity of PARG KO to PARGi through incomplete PARG depletion and demonstrate complete loss of PARG activity when incomplete PARG KO cells are transfected with additional gRNAs in the presence of PARPi. Furthermore, the

authors show that the sensitivity of PARG KO cells to PARGi is not caused by NAD depletion but by S-phase accumulation of pADPR on chromatin coming from unligated Okazaki fragments, which are recognized and bound by PARP1. Consistently, PARG KO or PARG inhibition show synthetic lethality with Pol beta, which is required for Okazaki fragment maturation. PARG expression levels in ovarian cancer cell lines correlate negatively with their sensitivity to PARGi.

Strengths:

The authors show that PARG is essential for removing ADP-ribosylation in S-phase.

Weaknesses:

1. This begs the question as to the relevant substrates of PARG in S-phase, which could be addressed, for example, by analysing PARylated proteins associated with replication forks in PARG-depleted cells (EdU pulldown and Af1521 enrichment followed by mass spectrometry).
2. The results showing the generation of a full PARG KO should be moved to the beginning of the Results section, right after the first Results chapter (PARG depletion leads to drastic sensitivity to PARGi), otherwise the reader is left to wonder how PARG KO cells can be sensitive to PARGi when there should be presumably no PARG present.
3. Please indicate in the first figure which isoforms were targeted with gRNAs, given that there are 5 PARG isoforms. You should also highlight that the PARG antibody only recognizes the largest isoform, which is clearly absent in your PARG KO, but other isoforms may still be produced, depending on where the cleavage sites were located.
4. FACS data need to be quantified. Scatter plots can be moved to Supplementary while quantification histograms with statistical analysis should be placed in the main figures.
5. All colony formation assays should be quantified and sensitivity plots should be shown next to example plates.
6. Please indicate how many times each experiment was performed independently and include statistical analysis.

- <https://doi.org/10.7554/eLife.89303.2.sa1>

Reviewer #3 (Public Review):

In the revised version the authors have addressed some of the reviewers' concerns, but, despite the new explanatory paragraph on page 16, the paper remains confusing because as shown in Figure 7 at the end of the Results the PARG KO 293A cells that were analyzed at the beginning of the Results are not true PARG knockouts. The authors stated that they did not rewrite the Results because they wanted to describe the experiments in the order in which they were carried out, but there is no imperative for the experiments to be described in the order in which they were done, and it would be much easier for the uninitiated reader to appreciate the significance of these studies if the true PARG KO cell data were presented at the beginning, as all three of the original reviewers proposed.

While the authors have to some extent clarified the nature of the PARG KO alleles, they have not been able to identify the source of the residual PARG activity in the PARG KO cells, in part because different commercial PARG antibodies give different and conflicting immunoblotting results. Additional sequence characterization of PARG mRNAs expressed in the PARG cKO cells, and also in-depth proteomic analysis of the different PARG bands could provide further insight into the origins and molecular identities of the various PARG proteins expressed from the different KO PARG alleles, and determine which of them might retain catalytic activity.

The authors have made no progress in identifying which are the key PARG substrates required for S phase progression, although they suggest that PARP1 itself may be an important target.

- <https://doi.org/10.7554/eLife.89303.2.sa0>

Author Response

The following is the authors' response to the original reviews.

eLife assessment

The authors' finding that PARG hydrolase removal of polyADP-ribose (PAR) protein adducts generated in response to the presence of unligated Okazaki fragments is important for S-phase progression is potentially valuable, but the evidence is incomplete, and identification of relevant PARylated PARG substrates in S-phase is needed to understand the role of PARylation and dePARylation in S-phase progression. Their observation that human ovarian cancer cells with low levels of PARG are more sensitive to a PARG inhibitor, presumably due to the accumulation of high levels of protein PARylation, suggests that low PARG protein levels could serve as a criterion to select ovarian cancer patients for treatment with a PARG inhibitor drug.

Thank you for the assessment and summary. Please see below for details as we have now addressed the deficiencies pointed out by the reviewers.

We believe that PARP1 is one of the major relevant PARG substrates in S phase cells. Previous studies reported that PARP1 recognizes unligated Okazaki fragments and induces S phase PARylation, which recruits single-strand break repair proteins such as XRCC1 and LIG3 that acts as a backup pathway for Okazaki fragment maturation (Hanzlikova et al., 2018; Kumamoto et al., 2021). In this study, we revealed that accumulation of PARP1/2-dependent S phase PARylation eventually led to cell death (Fig. 2). Furthermore, we found that chromatin-bound PARP1 as well as PARylated PARP1 increased in PARG KO cells (Fig. S4A and Fig. 4A), suggesting that PARP1 is one of the key substrates of PARG in S phase cells. Of course, PARG may have additional substrates besides PARP1 which are required for its roles in S phase progression, as PARG is known to be recruited to DNA damage sites through pADPr- and PCNA-dependent mechanisms (Mortusewicz et al., 2011). Precisely how PARG regulates S phase progression warrants further investigation.

Public Reviews:

Reviewer #1 (Public Review):

I have a major conceptual problem with this manuscript: How can the full deletion of a gene (PARG) sensitize a cell to further inhibition by its chemical inhibitor (PARGi) since the target protein is fully absent?

Please see below for details about this point. Briefly, we found that PARG is an essential gene (Fig. 7). There was residual PARG activity in our PARG KO cells, although the loss of full-length PARG was confirmed by Western blotting and DNA sequencing (Fig. S9). The residual PARG activity in these cells can be further inhibited by PARG inhibitor, which eventually lead to cell death.

The authors state in the discussion section: "The residual PARG dePARylation activity observed in PARG KO cells likely supports cell growth, which can be further inhibited by PARGi". What does this statement mean? Is the authors' conclusion that their PARG KOs

are not true KOs but partial hypomorphic knockdowns? Were the authors working with KO clones or CRISPR deletion in populations of cells?

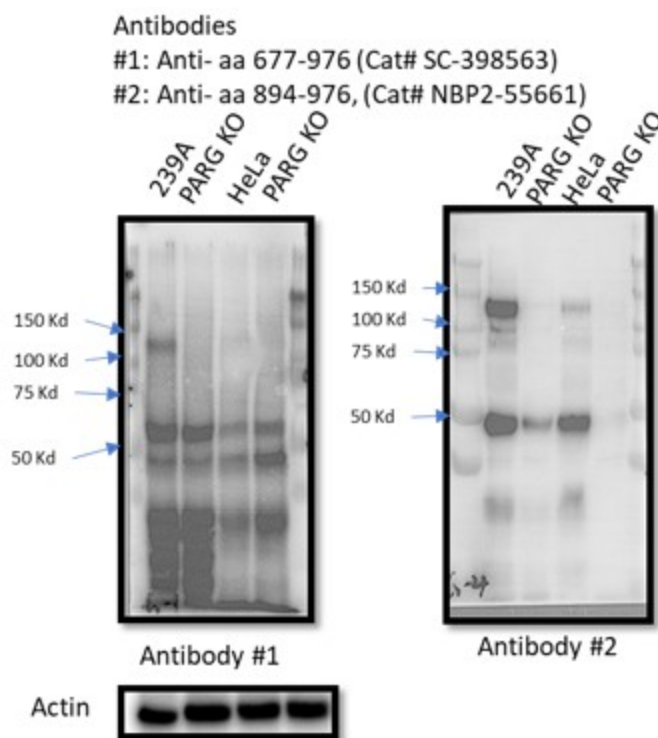
The reviewer is correct that our PARG KOs are not true KOs. We were working with CRISPR edited KO clones. As shown in this manuscript, we validated our KO clones by Western blotting, DNA sequencing and MMS-induced PARylation. Despite these efforts and our inability to detect full-length PARG in our KO clones, we suspect that our PARG KO cells may still express one or more active fragments of PARG due to alternative splicing and/or alternative ATG usage.

As shown in Fig. 7, we believe that PARG is essential for proliferation. Our initial KO cell lines are not complete PARG KO cells and residual PARG activity in these cells could support cell proliferation. Unfortunately, due to lack of appropriate reagents we could not draw solid conclusions regarding the isoforms or the truncated PARG expressed in these cells (Please see Western blots below).

Are there splice variants of PARG that were not knocked down? Are there PARP paralogues that can complement the biochemical activity of PARG in the PARG KOs? The authors do not discuss these critical issues nor engage with this problem.

There are five reviewed or potential PARG isoforms identified in the Uniprot database. The two sgRNAs (#1 and #2) used to generate initial PARG KO cells in this manuscript target all three catalytically active isoforms (isoforms 1, 2 and 3), and sgRNA#2 used in HeLa cells also targets isoforms 4 and 5, but these isoforms are considered catalytically inactive according to the Uniprot database. However, it is likely that sgRNA-mediated genome editing may lead to the creation of new alternatively spliced PARG mRNAs or the use of alternative ATG, which can produce catalytically active forms of PARG. Instead of searching for these putative spliced PARG RNAs, we used two independent antibodies that recognize the C-terminus of PARG for WB as shown below. Unfortunately, besides full-length PARG, these antibodies also recognized several other bands, some of them were reduced or absent in PARG KO cells, others were not. Thus, we could not draw a clear conclusion which functional isoform was expressed in our PARG KO cells. Nevertheless, we directly measured PARG activity in PARG KO cells (Fig. S9) and showed that we were still able to detect residual PARG activity in these PARG KO cells. These data clearly indicate that residual PARG activity are present and detected in our KO cells, but the precise nature of these truncated forms of PARG remains elusive.

Author response image 1.



These issues have to be dealt with upfront in the manuscript for the reader to make sense of their work.

We thank this reviewer for his/her constructive comments and suggestions. We will include the data above and additional discussion upfront in our revised manuscript to avoid any further confusion by our readers.

Reviewer #2 (Public Review):

Summary:

In this manuscript, Nie et al investigate the effect of PARG KO and PARG inhibition (PARGi) on pADPR, DNA damage, cell viability, and synthetic lethal interactions in HEK293A and HeLa cells. Surprisingly, the authors report that PARG KO cells are sensitive to PARGi and show higher pADPR levels than PARG KO cells, which are abrogated upon deletion or inhibition of PARP1/PARP2. The authors explain the sensitivity of PARG KO to PARGi through incomplete PARG depletion and demonstrate complete loss of PARG activity when incomplete PARG KO cells are transfected with additional gRNAs in the presence of PARPi. Furthermore, the authors show that the sensitivity of PARG KO cells to PARGi is not caused by NAD depletion but by S-phase accumulation of pADPR on chromatin coming from unligated Okazaki fragments, which are recognized and bound by PARP1. Consistently, PARG KO or PARG inhibition shows synthetic lethality with Pol beta, which is required for Okazaki fragment maturation. PARG expression levels in ovarian cancer cell lines correlate negatively with their sensitivity to PARGi.

Thank you for your nice comments. The complete loss of PARG activity was observed in PARG complete/conditional KO (cKO) cells. These cKO clones were generated using wild-type cells

transfected with sgRNAs targeting the catalytic domain of PARG in the presence of PARP inhibitor.

Strengths:

The authors show that PARG is essential for removing ADP-ribosylation in S-phase.

Thanks!

Weaknesses:

- 1. This begs the question as to the relevant substrates of PARG in S-phase, which could be addressed, for example, by analysing PARylated proteins associated with replication forks in PARG-depleted cells (EdU pulldown and Af1521 enrichment followed by mass spectrometry).*

We believe that PARP1 is one of the major relevant PARG substrates in S phase cells. Previous studies reported that PARP1 recognizes unligated Okazaki fragments and induces S phase PARylation, which recruits single-strand break repair proteins such as XRCC1 and LIG3 that acts as a backup pathway for Okazaki fragment maturation (Hanzlikova et al., 2018; Kumamoto et al., 2021). In this study, we revealed that accumulation of PARP1/2-dependent S phase PARylation eventually led to cell death (Fig. 2). Furthermore, we found that chromatin-bound PARP1 as well as PARylated PARP1 increased in PARG KO cells (Fig. S4A and Fig. 4A), suggesting that PARP1 is one of the key substrates of PARG in S phase cells. Of course, PARG may have additional substrates besides PARP1 which are required for its roles in S phase progression, as PARG is known to be recruited to DNA damage sites through pADPr- and PCNA-dependent mechanisms (Mortusewicz et al., 2011). Precisely how PARG regulates S phase progression warrants further investigation.

- 1. The results showing the generation of a full PARG KO should be moved to the beginning of the Results section, right after the first Results chapter (PARG depletion leads to drastic sensitivity to PARGi), otherwise, the reader is left to wonder how PARG KO cells can be sensitive to PARGi when there should be presumably no PARG present.*

Thank you for your suggestion! However, we would like to keep the complete PARG KO result at the end of the Results section, since this was how this project evolved. Initially, we did not know that PARG is an essential gene. Thus, we speculated that PARGi may target not only PARG but also a second target, which only becomes essential in the absence of PARG. To test this possibility, we performed FACS-based and cell survival-based whole-genome CRISPR screens (Fig. 5). However, this putative second target was not revealed by our CRISPR screening data (Fig. 5). We then tested the possibility that these cells may have residual PARG expression or activity and only cells with very low PARG expression are sensitive to PARGi, which turned out to be the case for ovarian cancer cells. Equipped with PARP inhibitor and sgRNAs targeting the catalytic domain of PARG, we finally generated cells with complete loss of PARG activity to prove that PARG is an essential gene (Fig. 7). This series of experiments underscore the challenge of validating any KO cell lines, i.e. the identification of frame-shift mutations, absence of full-length proteins, and phenotypic changes may still not be sufficient to validate KO clones. This is an important lesson we learned and we would like to share it with the scientific community.

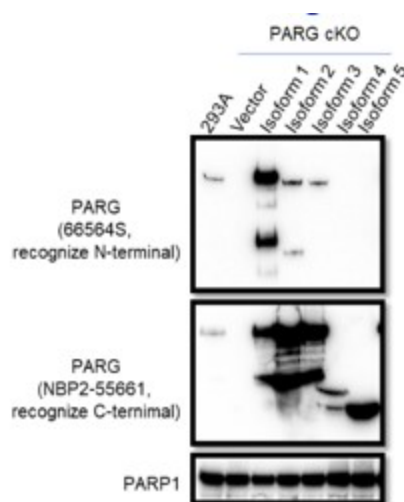
To avoid further misunderstanding, we will include additional statements/comments at the end of “PARG depletion leads to drastic sensitivity to PARGi” section and at the beginning of “CRISPR screens reveal genes responsible for regulating pADPr signaling and/or cell lethality in WT and PARG KO cells”. Hope that our revised manuscript will make it clear.

1. Please indicate in the first figure which isoforms were targeted with gRNAs, given that there are 5 PARG isoforms. You should also highlight that the PARG antibody only recognizes the largest isoform, which is clearly absent in your PARG KO, but other isoforms may still be produced, depending on where the cleavage sites were located.

The two sgRNAs (#1 and #2) used to generate initial PARG KO cells in this manuscript target all three catalytically active isoforms (isoforms 1, 2 and 3), and sgRNA#2 used in HeLa cells also targets isoforms 4 and 5, but these isoforms are considered catalytically inactive according to the Uniprot database. As suggested, we will modify Fig. S1D and the figure legends.

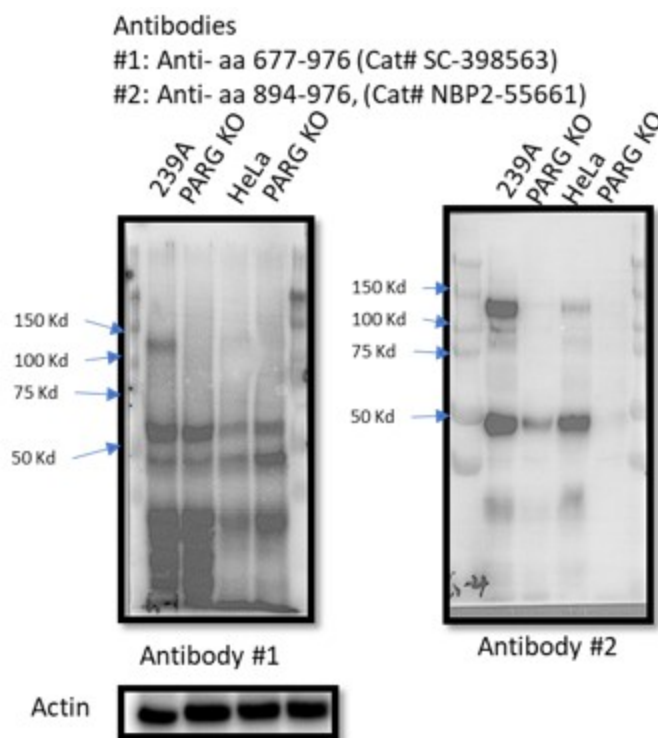
The manufacturer instruction states that the Anti-PARG antibody (66564S) can only recognize isoform 1, this antibody could recognize isoforms 2 and 3 albeit weakly based on Western blot results with lysates prepared from PARG cKO cells reconstituted with different PARG isoforms, as shown below. As suggested, we will add a statement in the revised manuscript and provide the Western blotting data below.

Author response image 2.



To test whether other isoforms were expressed in 293A and/or HeLa cells, we used two independent antibodies that recognize the C-terminus of PARG for WB as shown below. Unfortunately, besides full-length PARG, these antibodies also recognized several other bands, some of them were reduced or absent in PARG KO cells, others were not. Thus, we could not draw a clear conclusion which functional isoforms or truncated forms were expressed in our PARG KO cells.

Author response image 3.



1. FACS data need to be quantified. Scatter plots can be moved to Supplementary while quantification histograms with statistical analysis should be placed in the main figures.

We agree with this reviewer that quantification of FACS data may provide straightforward results in some of our data. However, it is challenging to quantify positive S phase pADPr signaling in some panels, for example in Fig. 3A and Fig. 4C. In both panels, pADPr signaling was detected throughout the cell cycle and therefore it is difficult to know the percentage of S phase pADPr signaling in these samples. Thus, we decide to keep the scatter plots to demonstrate the dramatic and S phase-specific pADPr signaling in PARG KO cells treated with PARGi. We hope that these data are clear and convincing even without any quantification.

1. All colony formation assays should be quantified and sensitivity plots should be shown next to example plates.

As suggested, we will include the sensitivity plot next to Fig. 3D. However, other colony formation assays in this study were performed with a single concentration of inhibitor and therefore we will not provide sensitivity plots for these experiments. Nevertheless, the results of these experiments are straightforward and easy to interpret.

1. Please indicate how many times each experiment was performed independently and include statistical analysis.

As suggested, we will add this information in the revised manuscript.

Reviewer #3 (Public Review):

Here the authors carried out a CRISPR/sgrRNA screen with a DDR gene-targeted mini-library in HEK293A cells looking for genes whose loss increased sensitivity to treatment with the PARG inhibitor, PDD00017273 (PARGi). Surprisingly they found that PARG itself, which encodes the cellular poly(ADP-ribose) glycohydrolase (dePARylation) enzyme, was a major hit. Targeted PARG KO in 293A and HeLa cells also caused high sensitivity to PARGi. When PARG KO cells were reconstituted with catalytically-dead PARG, MMS treatment caused an increase in PARylation, not observed when cells were reconstituted with WT PARG or when the PARG KO was combined with PARP1/2 DKO, suggesting that loss of PARG leads to a strong PARP1/2-dependent increase in protein PARylation. The decrease in intracellular NADH⁺, the substrate for PARP-driven PARylation, observed in PARG KO cells was reversed by treatment with NMN or NAM, and this treatment partially rescued the PARG KO cell lethality. However, since NAD⁺ depletion with the FK868 nicotinamide phosphoribosyltransferase (NAMPT) inhibitor did not induce a similar lethality the authors concluded that NAD⁺ depletion/reduction was only partially responsible for the PARGi toxicity. Interestingly, PARylation was also observed in untreated PARG KO cells, specifically in S phase, without a significant rise in γH2AX signals. Using cells synchronized at G1/S by double thymidine blockade and release, they showed that entry into S phase was necessary for PARGi to induce PARylation in PARG KO cells. They found an increased association of PARP1 with a chromatin fraction in PARG KO cells independent of PARGi treatment, and suggested that PARP1 trapping on chromatin might account in part for the increased PARGi sensitivity. They also showed that prolonged PARGi treatment of PARG KO cells caused S phase accumulation of pADPr eventually leading to DNA damage, as evidenced by increased anti-γH2AX antibody signals and alkaline comet assays. Based on the use of emetine, they deduced that this response could be caused by unligated Okazaki fragments. Next, they carried out FACS-based CRISPR screens to identify genes that might be involved in cell lethality in WT and PARG KO cells, finding that loss of base excision repair (BER) and DNA repair genes led to increased PARylation and PARGi sensitivity, whereas loss of PARP1 had the opposite effects. They also found that BER pathway disruption exhibited synthetic lethality with PARGi treatment in both PARG KO cells and WT cells, and that loss of genes involved in Okazaki fragment ligation induced S phase pADPr signaling. In a panel of human ovarian cancer cell lines, PARGi sensitivity was found to correlate with low levels of PARG mRNA, and they showed that the PARGi sensitivity of cells could be reduced by PARPi treatment. Finally, they addressed the conundrum of why PARG KO cells should be sensitive to a specific PARG inhibitor if there is no PARG to inhibit and found that the PARG KO cells had significant residual PARG activity when measured in a lysate activity assay, which could be inhibited by PARGi, although the inhibited PARG activity levels remained higher than those of PARG cKO cells (see below). This led them to generate new, more complete PARG KO cells they called complete/conditional KO (cKO), whose survival required the inclusion of the olaparib PARPi in the growth medium. These PARG cKO cells exhibited extremely low levels of PARG activity in vitro, consistent with a true PARG KO phenotype.

We thank this reviewer for his/her constructive comments and suggestions.

The finding that human ovarian cancer cells with low levels of PARG are more sensitive to inhibition with a small molecule PARG inhibitor, presumably due to the accumulation of high levels of protein PARylation (pADPr) that are toxic to cells is quite interesting, and this could be useful in the future as a diagnostic marker for preselection of ovarian cancer patients for treatment with a PARG inhibitor drug. The finding that loss of base excision repair (BER) and DNA repair genes led to increased PARylation and PARGi sensitivity is in keeping with the conclusion that PARG activity is essential for cell fitness,

because it prevents excessive protein PARylation. The observation that increased PARylation can be detected in an unperturbed S phase in PARG KO cells is also of interest. However, the functional importance of protein PARylation at the replication fork in the normal cell cycle was not fully investigated, and none of the key PARylation targets for PARG required for S phase progression were identified. Overall, there are some interesting findings in the paper, but their impact is significantly lessened by the confusing way in which the paper has been organized and written, and this needs to be rectified.

We believe that PARP1 is one of the major relevant PARG substrates in S phase cells. Previous studies reported that PARP1 recognizes unligated Okazaki fragments and induces S phase PARylation, which recruits single-strand break repair proteins such as XRCC1 and LIG3 that acts as a backup pathway for Okazaki fragment maturation (Hanzlikova et al., 2018; Kumamoto et al., 2021). In this study, we revealed that accumulation of PARP1/2-dependent S phase PARylation eventually led to cell death (Fig. 2). Furthermore, we found that chromatin-bound PARP1 as well as PARylated PARP1 increased in PARG KO cells (Fig. S4A and Fig. 4A), suggesting that PARP1 is one of the key substrates of PARG in S phase cells. Of course, PARG may have additional substrates besides PARP1 which are required for its roles in S phase progression, as PARG is known to be recruited to DNA damage sites through pADPr- and PCNA-dependent mechanisms (Mortusewicz et al., 2011). Precisely how PARG regulates S phase progression warrants further investigation.

As suggested, we will revise our manuscript accordingly and provide additional explanation/statement upfront to avoid any misunderstandings.

Reviewer #1 (Recommendations For The Authors):

1. Figure 1c. Why does the viability of PARG KO cells improve at higher doses of PARGi? How do the authors explain this paradox?

This phenomenon was observed in 293A PARG KO cells and happened in CellTiter-Glo assay, especially with the top three PARGi concentrations (100 μ M, 33.33 μ M and 11.11 μ M). This may due to the low solubility of this PARGi in the medium, since we sometimes observed precipitation at high concentrations when PARGi stock was diluted in medium.

1. Figure 2d. The authors show that PARGi reduced NAD⁺ level by 20%. This reduction in NAD⁺ probably does not explain the cell death phenotype observed by parthanatos cell death. What pathway is activated by PARGi to induce cell death?

Since PARG KO cells treated with PARGi led to uncontrolled pADPr accumulation, it is possible that some of these cells may die due to parthanotos. However, we did not observe a dramatic reduction in NAD⁺ level. A previous study showed that Parg(-/-) mouse ES cells predominantly underwent caspase-dependent apoptosis (Shirai et al., 2013). Indeed, PARP1 cleavage was detected in PARG KO cells with prolonged PARGi treatment, indicating that at least some of these cells die due to apoptosis (Fig. 2A). Cytotoxicity of PARGi in PARG KO cells may due to several mechanisms including apoptosis, parthanatos and NAD⁺ reduction.

1. The authors refer to FK866 in the text without explaining what this agent is. FK866 is a noncompetitive inhibitor of nicotinamide phosphoribosyltransferase (NAPRT), a key enzyme in the regulation of NAD⁺ biosynthesis from the natural precursor nicotinamide. The authors should explain experimental tools in the text as they use them for clarity to the reader.

Thanks for the suggestion! We will include additional citations and discuss how FK866 works in our revised manuscript.

1. *In addition to these issues, there are significant formatting and textual problems, such that there are multiple gaps in the body of the text that make coherent reading of the manuscript impossible. Examples are: Page 3 line 10. Page 6 line 5 and line 15, Page 7 line 2, 3, and line 8. Page 8, line 1, and line 3 from bottom. Page 9 line 1, line 7 from bottom and line 9 from the bottom, Page 18 of the results in several places, etc. etc. These formatting errors convey the impression that the submitting authors did not adequately review the manuscript for technical problems prior to submission. The authors need to correct these errors.*

Sorry, we will edit the text and remove these gaps as suggested.

Reviewer #3 (Recommendations For The Authors):

1. *The major problem with this paper is conceptual - namely, how could PARG knockout cells be hypersensitive to a selective PARG small molecular inhibitor. The evidence in Figure 7 that there is measurable residual PARG activity in the so-called PARG KO 293A and HeLa cells provides a partial explanation for why PARG inhibitor treatment might be deleterious to the PARG KO cells, i.e., because PARGi blocks this residual PARG activity. However, although the authors characterized the PARG alleles in the 293A PARG KO cells by sequencing, the molecular origin of the significant level of residual PARG activity remains unclear (see points 7-9).*

Yes, in our study we showed that PARGi treatment inhibited the residual PARG activity in PARG KO cells, which mimics complete loss of PARG as PARG is an essential gene. These data agree with a previous study using Parg(-/-) mouse cells (Koh et al., 2004). We attempted to define the molecular origin of the residual PARG activity, unfortunately this was challenging (please see below for additional discussions). Nevertheless, we showed that residual PARG activity could be detected in PARG KO cells and more importantly cells with reduced PARG expression or activity are sensitive to PARGi. These results indicate that PARG expression and/or activity may be used as a biomarker for PARGi-based therapy.

1. *Although the most obvious explanation for the PARGi sensitivity data presented in Figures 1-4 is that the PARG KO cells have residual PARG activity, the authors wait until the discussion on page 26 to raise the possibility that the PARG KO cells might have residual PARG activity that renders them sensitive to PARGi. It would be more logical to move the PARG activity data in Figure 7 earlier in the paper as a supplementary figure, so that the reader is not left wondering how a PARG KO cell remains sensitive to a PARG inhibitor. For this reason, it is recommended that the whole paper be reorganized and rewritten to provide a more logical flow that allows the reader to understand what was done, and why it is hard to generate complete PARG KO cells because the accumulation of pADPR adducts is toxic to the cell.*

Thank you for your suggestion! However, we would like to keep the complete PARG KO result at the end of the Results section, since this was how this project evolved. Initially, we did not know that PARG is an essential gene. Thus, we speculated that PARGi may target not only PARG but also a second target, which only becomes essential in the absence of PARG. To test this possibility, we performed FACS-based and cell survival-based whole-genome CRISPR screens (Fig. 5). However, this putative second target was not revealed by our CRISPR screening data (Fig. 5). We then tested the possibility that these cells may have residual PARG

expression or activity and only cells with very low PARG expression are sensitive to PARGi, which turned out to be the case for ovarian cancer cells. Equipped with PARP inhibitor and sgRNAs targeting the catalytic domain of PARG, we finally generated cells with complete loss of PARG activity to prove that PARG is an essential gene (Fig. 7). This series of experiments underscore the challenge of validating any KO cell lines, i.e. the identification of frame-shift mutations, absence of full-length proteins, and phenotypic changes may still not be sufficient to validate KO clones. This is an important lesson we learned and we would like to share it with the scientific community.

To avoid further misunderstanding, we will include additional statements/comments at the end of “PARG depletion leads to drastic sensitivity to PARGi” section and at the beginning of “CRISPR screens reveal genes responsible for regulating pADPr signaling and/or cell lethality in WT and PARG KO cells”. Hope that our revised manuscript will make it clear.

1. Exactly how PARG activity would be coordinated with PARP1/2 activity during normal S phase to ensure that PARylation can serve its required function, whatever that may be, and is then removed by PARG is unclear - how would this be orchestrated at the level of a replication fork?

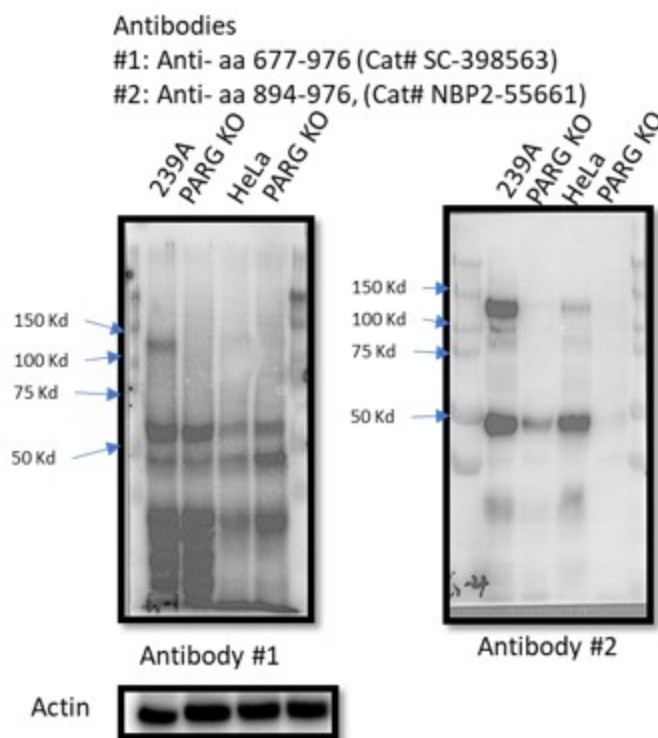
PARG is known to be recruited to sites of DNA damage through pADPr- and PCNA-dependent mechanisms (Mortusewicz et al., 2011). Our current hypothesis is that PARP1 is one of the major PARG substrates in S phase cells. Previous studies reported that PARP1 recognizes unligated Okazaki fragments and induces S phase PARylation, which recruits single-strand break repair proteins such as XRCC1 and LIG3 that acts as a backup pathway for Okazaki fragment maturation (Hanzlikova et al., 2018; Kumamoto et al., 2021). In this study, we revealed that accumulation of PARP1/2-dependent S phase PARylation eventually led to cell death (Fig. 2). Furthermore, we found that chromatin-bound PARP1 as well as PARylated PARP1 increased in PARG KO cells (Fig. S4A and Fig. 4A), suggesting that PARP1 is one of the key substrates of PARG in S phase cells. Of course, PARG may have additional substrates besides PARP1 which are required for its roles in S phase progression. Precisely how PARG regulates S phase progression warrants further investigation.

1. Figure 2B: What gRNAs were used to generate the 293A and HeLa PARG knock clones, i.e., where are they located in the PARG gene? If they are not in the catalytic domain it might be possible to generate PARG proteins with N-terminal deletions that are still active (see points 8-10 below).

The two sgRNAs (#1 and #2) used to generate initial PARG KO cells in this manuscript target all three catalytically active isoforms (isoforms 1, 2 and 3), and sgRNA#2 used in HeLa cells also targets isoforms 4 and 5, but these isoforms are considered catalytically inactive according to the Uniprot database. As suggested, we will modify Fig. S1D and the figure legends to show the localization of gRNAs.

We agree with this reviewer that truncated but active forms of PARG exist in these KO cells. We attempted to identify these truncated forms of PARG by using two independent antibodies that recognize the C-terminus of PARG for WB as shown below. Unfortunately, besides full-length PARG, these antibodies also recognized several other bands, some of them were reduced or absent in PARG KO cells, others were not. Thus, we could not draw a clear conclusion which functional isoform/truncated form was expressed in our PARG KO cells. Nevertheless, we directly measured PARG activity in PARG KO cells (Fig. S9) and showed that we were still able to detect residual PARG activity in these PARG KO cells. Based on these results, we stated that the residual PARG activity was detected in our KO cells, but we were not able to specify the truncated variants of PARG in these cells.

Author response image 4.



1. Figure 3B/page 19: The authors state that "emetine, which diminishes Okazaki fragments, greatly inhibited S phase pADPr signaling in PARG KO cells", and from this deduced that Okazaki fragments on the lagging strand activate PARylation. However, emetine is not a specific lagging strand synthesis inhibitor, as implied here, but rather a protein synthesis inhibitor, which inhibits Okazaki fragment formation indirectly (see PMID: 36260751). The authors need to rewrite this section to explain how emetine works in this context.

As suggested, we will cite this reference and discuss how emetine inhibits Okazaki fragment maturation in our revised manuscript. Additionally, we used three different POLA1 inhibitors to diminish Okazaki fragments. As shown in Fig. S3B, all three POLA1 inhibitors significantly abolished S-phase pADPr induced by PARGi in PARG KO cells. Furthermore, POLA1 inhibitors, adarotene and CD437, were able to rescue cell lethality caused by PARGi in PARG KO cells (Fig. 3E).

1. Figure 7: It is not clear why these cells are called PARG complete/conditional KO cells (cKO). Generally, "conditional knockout" refers to a cell or animal in which a gene can be conditionally knocked out by inducible expression of Cre. Here, it appears that "conditional" refers to the fact that the PARG KO cells only grow in the presence of olaparib - is this the case?

Yes, we used the name to separate these cells from our initial PARG KO cells. Moreover, we were only able to obtain and maintain these PARG cKO clones with complete loss of PARG activity in the presence of PARP inhibitor. Therefore, we called them PARG complete/conditional KO (cKO) cells.

1. Figure 7B and D: The level of full-length PARG protein was much lower in the 293A and HeLa cKO cells compared to WT cells consistent with cKO cells representing a more complete PARG KO. The level of PARG protein in the 293A PARG cKO cells was apparently also lower than in the original PARG KO cells, but the KO and cKO samples should be run side by side to demonstrate this conclusively, and the bands need to be quantified. In panel B, it is not clear from the legend what cKO_3 and cKO_4 are, but presumably, they are different clones, and this should be stated.

Full-length PARG was not detected in either PARG KO or PARG cKO cells by WB. The apparent lower level of endogenous PARG in Fig. 7D was due to the fact that reconstituted cells had high exogenous PARG expression and therefore we had to reduce exposure time for WB.

As for cKO_3 and cKO_4 in Fig.7, they are different clones created by different sgRNAs. As suggested, we will include additional information in figure legends to clearly state which sgRNA was used to generate the respective KO and cKO clones.

1. Figure S8: There is not enough information here or in the text to allow the reader to interpret these PARG allele sequences obtained from the PARG KO cells. From the Methods section, it appears that the PARG KO cells were clonal, with sequence data from one clone of each of the 293A and HeLa cell PARG KO cells being shown. If this is right, then in both cell types one out of four PARG alleles is wild type, and therefore one would expect the PARG protein signal to be ~25% of that in WT cells. However, based on the 293A PARG KO cells PARG immunoblot in Figure 2B the PARG protein signal is clearly much lower than 25% (these bands need to be quantified), and this discrepancy needs to be explained. What is the level of PARG protein in the PARG KO HeLa cells? If different PARG KO cell clones are analyzed by sequencing, do they all have an apparently intact PARG allele? Four different gRNA target sites in the PARG gene are shown in panel A in Figure 7, but the description in the text regarding how the four gRNAs were used is totally inadequate - were all four used simultaneously or only the two in the catalytic domain? Were pairs of gRNAs used in an attempt to generate a large intervening deletion - some Southern blots of the PARG gene region in the PARG cKO cells are needed to figure this out. The gRNAs are given numbers in Figure 7A, but it is unclear from the sequences shown in Figures S8 and S9 which gRNA sites are shown. All of this has to be clarified, so that the reader can understand the nature of the KO/cKO cells knockout alleles, and what PARG-related products, if any, they can express.

Yes, all KO and cKO cells used in this study are single clones. As suggested, we will revise figure legends in Fig.7, S8 and S9 to include detailed information. To avoid any further misunderstanding, we will label the allele “WT” to “WT (reference)” in Fig. S8 and S9. We did not detect intact/wild-type PARG sequence in any single KO/cKO clone by DNA sequencing. Sequencing of single KO/cKO clones was performed by using TOP TA Cloning kit. Briefly, genomic DNA was extracted from each single KO/cKO clone. Approximately 300bp surrounding the sgRNA targeting sequence was amplified by PCR. The PCR product was cloned into the vector and approximately 10-15 bacteria clones were extracted and sent for sequencing. If any intact/wild-type PARG sequence was detected in these 10-15 bacteria clones, this KO/cKO clone was considered heterozygous clone and discarded.

HEK293A and HeLa cells are not diploid cells and have complex karyotypes. PARG gene is located on chromosome 10. Karyotyping by M-FISH shows that HeLa cells have 3 copies of chromosome 10 (Landry et al., 2013). HEK293 cells predominantly have 3 copies of chromosome 10 and sometimes 4 copies can be detected by G-banding (Binz et al., 2019). Therefore, it is anticipated that 1 to 4 mutant alleles would be detected in each KO/cKO clone by sequencing.

Only one sgRNA was transfected into cells for the selection of single clones. We did not use paired or multiple sgRNAs in any of these experiments. As shown in Fig. S1D and Fig. 7A, HEK293A derived and HeLa derived PARG KO single clones were generated with the use of different sgRNAs. In addition, the two PARG cKO single clones from HEK293A and HeLa cells were also generated by the use of two different sgRNAs, as shown in Fig. 7A-B. We will include all the information above in the revised manuscript, i.e. in Methods section as well as in figure legends.

1. *Figure S9A: The sequences of the 293A PARG alleles in the cKO cells suggest that these cells also have one intact PARG allele, which again does not fit with the very low level of intact PARG protein shown in Figure 7B. How do the authors explain this?*

Sorry, this is a misunderstanding. The allele “WT” in Fig. S8 and S9 is the reference sequence. We will change it to “Reference sequence” to avoid further confusion. As mentioned above, we did not detect any intact/wild-type PARG sequence in any of our single KO/cKO clones by sequencing.

1. *Figure S9B: These critical lysate activity data show that the PARG KO cells have ~50% of the PARG activity detected in WT cells. However, this is not consistent with the PARG protein level detected in PARG immunoblot in Figure 1B, which appears to be less than 5% of the PARG protein level in WT cells (with one intact PARG allele in these cells one would theoretically expect ~25%, although this depends on whether all four alleles are expressed equally). One possibility is that active PARG fragments are generated from one or more of the PARG KO alleles in the PARG KO cells. Targeted sequencing of PARG mRNAs might reveal whether there are shorter RNAs that could encode a protein containing the C-terminal catalytic domain (aa 570-910). In addition, the authors need to show the entire immunoblot to determine if there are smaller proteins recognized by the anti-PARG antibodies that might represent shorter PARG gene products (for this we need to know where the epitope against which the PARG antibodies are directed are located within the PARG protein - ideally they authors need to use an antibody directed against an epitope near the C-terminus).*

As stated in the Methods section, we incubated cell lysates with substrates overnight to evaluate the maximum level of pADPr hydrolysis, i.e. PARG activity, we were able to detect in this assay. It is very likely that the PARG activity in PARG KO cells was much lower than 50%, due to saturation of signals for lysates isolated from wild-type cells. Thus, the data presented in our manuscript probably underestimate the reduction of PARG activity in PARG KO cells. Nevertheless, these data indicate that residual PARG activity was detected in PARG KO cells, however this activity was absent in PARG cKO cells.

As aforementioned, we used two independent antibodies that recognize the C-terminus of PARG for WB. Unfortunately, we could not draw a clear conclusion which functional isoforms or truncated proteins were expressed in our PARG KO cells. The dePARylation assay used here may be the best way to test the residual PARG activity in our KO and cKO cells.

1. *Figure 7D: In this experiment, the level of re-expressed WT PARG protein was much higher than that of the endogenous PARG protein (quantification is needed) - how might this affect the interpretation of these experiments (N.B., WT and catalytically-dead PARG were also re-expressed for the experiments shown in Figure 1, but there are no PARG immunoblots to demonstrate how much the exogenous proteins were overexpressed, or activity measurements). If regulated pADPr signaling is important for a normal S phase, then one would have thought that expressing a very high level of active PARG would create problems.*

In Fig. S1E, we blotted endogenous PARG level in control cells and exogenous PARG level in reconstituted cells. The reviewer is correct that exogenous PARG expression was much higher (~10-fold) than that of endogenous PARG in WT control cells. Nevertheless, we did not observe any obvious phenotypes in PARG KO/cKO cells reconstituted with high level of exogenous PARG, which may reflect excess PARG level/activity in wild-type control cells.

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